

ICOHP Descriptors and Thermodynamic Stability Across a 597-Structure Dataset

Campaign 2: Dataset Construction, VASP+LOBSTER Computation, and Statistical Analysis

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The project’s central question is viability discrimination based on $\Delta(\text{ICOHP})$ (§6.2), across a 597-structure dataset.

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1 Objective and Problem Statement

Central research question (see §6.2): does $\Delta(\text{ICOHP})$ — the reaction-ICOHP descriptor for a compound’s decomposition into elements, and specifically the sign-based endobondic/exobondic classification it supports — distinguish thermodynamically stable, metastable, and unstable compounds?

2 Methodology

2.1 Dataset Selection

Materials Project query: unary and binary chemical systems, non-magnetic ($|\text{total magnetization}| \leq 0.01 \mu_B/\text{formula unit}$ — MP’s categorical `magnetic_ordering` field is "Unknown" for $\sim 98\%$ of entries and would have excluded nearly every genuinely non-magnetic compound), no f-block elements. The dataset combines two complementary selection tracks: a first set of 186 compounds (≤ 20 atoms/cell, 60 per stability family, a 2-entries-per-chemical-system cap to ensure elemental diversity rather than many polymorphs of the same system), and a second set of 163 compounds/elements (≤ 40 atoms/cell) specifically targeting additional elemental references, off-equilibrium polymorphs (high-pressure phases, allotropes), and alkali/alkaline-earth binaries against N/O/F/P/S/Cl, chosen to build genuine same-composition polymorph groups and to test descriptors far from equilibrium; and a third set of 45 binary compounds, each pairing an experimentally known, maximally metastable compound (farthest above the convex hull among its chemical system) with 1–3 theoretical polymorphs of the same formula sitting even farther above the hull, a robustness test deliberately targeting stability edge cases.

2.2 Bonding-Type Classification

Bond type (ionic / covalent / metallic) is computed by default by the `classify()` function (elemental composition + `is_metal`). A second, independent classification is built directly from each compound’s ICOHP/ICOB data (§5): `is_metal` for metallic character, then the mean ICOBI per bond (bond order, covalency) to separate ionic from covalent among non-metallic compounds. The two classifications are compared in §5.

2.3 VASP + LOBSTER Computation

Same template across all structures (static run, `ISYM=-1`, `LASPH=.TRUE.`, PAW-PBE potentials recommended by `pymatgen.io.vasp.sets.MPRelaxSet`): NBANDS is derived per compound from the number of local orbitals in the LOBSTER basis (`Lobsterin.write_INCAR`), and basis-function determination goes through `Lobsterin.get_basis(potcar_symbols=...)` rather than direct POTCAR-file parsing (whose internal pymatgen validity check fails on several POTCARs unrecognized by hash in the local mirror).

2.4 ICOHP/ICOB Quantities: Integration, Cutoffs, Units

LOBSTER generation (`lobsterin`, identical across the dataset): `pbeVaspFit2015` basis; COHP integration window from -35 to $+5$ eV around the Fermi level (`COHPstartEnergy/COHPendEnergy`); bond detection by interatomic distance from 0.1 to 6.0 Å (`cohpGenerator`, orbital-resolved); 0.05 eV Gaussian smearing (`gaussianSmearingWidth`).

Integrated ICOHP/ICOB (`ICOHPLIST.lobster/ICOBILIST.lobster`). A bond’s integrated ICOHP is, by LOBSTER’s own definition, the integral of $\text{COHP}(E)$ from the bottom of the window above up to the Fermi level (occupied states only), in eV; negative = bonding, positive = antibonding. ICOBI is the same integral over COBI (bond order index), dimensionless.

Per-compound aggregation (`percolation_path.py`): `icohp_sum/icobi_sum` over every symmetrically inequivalent bond label in `ICOHPLIST.lobster/ICOBILIST.lobster` (each periodic bond counted once, not once per direction); `icohp_mean/icobi_mean`, the corresponding per-bond average — this intensive version, comparable across compounds of different cell sizes, is what is reported in Table 10 and the “ICOHP”/“ICOB” columns of Appendix A. These quantities feed §5 and the classic correlations of §6.

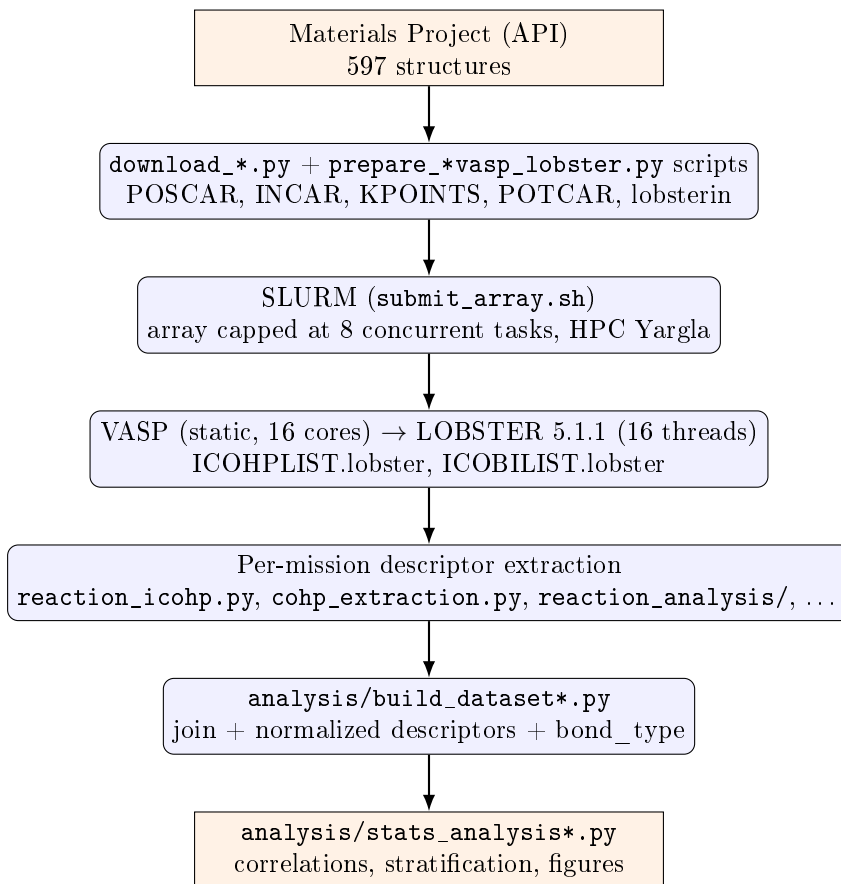
Reaction $\Delta(\text{ICOHP})$ (`reaction_analysis/`). For a balanced reaction (typically compound $\rightarrow$ elements), Δ = sum of products’ ICOHP – sum of reactants’ ICOHP, "products – reactants" convention, in three normalizations: per formula unit (extensive), per transferred atom (the normalization used throughout this report unless noted), and per bond (diagnostic only, non-conservative — bonds are not conserved between differently-bonded structures). Positive sign ("endobondic"): breaking the reactant’s bonds costs more than the products’ bonds recover — a possible kinetic barrier. Negative sign ("exobondic"): no such barrier is visible from bonding alone.

Antibonding population near the Fermi level (`cohpc_extraction.py`) — a quantity distinct from the $\Delta(\text{ICOHP})$ above. Rather than integrating the whole occupied COHP, this quantity integrates only its antibonding part ($\text{COHP} > 0$) over a one-sided window ($E_{\text{ref}} - \Delta E, E_{\text{ref}}$], where $E_{\text{ref}} = E_F$ (metallic compounds) or the VBM (gapped compounds), and $\Delta E \in \{0.5, 1.0, 2.0\}$ eV ($\Delta E = 1.0$ eV as the primary window) — only occupied states can destabilize the ground state via antibonding, hence the one-sided rather than symmetric window. Reported both raw (same "eV" unit convention as LOBSTER, though it is a Hamilton population integral, not literally an energy) and normalized by the total occupied ICOHP magnitude at the same E_{ref} . This quantity, and only this one, feeds the descriptor in §6.2 — the reaction $\Delta(\text{ICOHP})$ (§6.2–6.2) integrates the whole occupied COHP with no restriction to a window near E_F .

2.5 Statistics

Spearman correlation (no reason to expect a linear relationship) between each ICOHP/ICOB descriptor and a stability target (`energy_above_hull` or `formation_energy_per_atom` depending on the descriptor), overall and per subgroup (`bond_type`, `is_metal`); any $n < 15$ subgroup is explicitly flagged, never presented bare. **No SISSO, no symbolic regression** at this stage (see §9).

3 Workflow Diagram



4 Environment

Identical across all 597 structures (Yargla, VASP 6.5.0, LOBSTER 5.1.1, Python 3.11 `.venv`), extended with: `mp-api`, `pandas`, `scipy`, `scikit-learn`, `matplotlib`, `pydantic`, `pyyaml` (see `requirements.txt`). Per-step core allocation: §8.

5 Dataset

The full set of computed compounds and elements totals **597 structures**.

Table 1: Dataset composition by element count.

Composition	<i>n</i>
Single-element references	69
Multi-element compounds	522
Validation structures without compound metadata*	6
Total	597

* isolated gas dimers (H_2 , N_2 , O_2 , F_2 , Cl_2) and $\text{CaO}[\text{sphalerite}]$, used only for the Reitz & Dronskowski validation (Appendix K), outside the main dataset.

Bond type. Two independent classifications exist (§2.2) — `classify()` and a classification built directly from ICOHP/ICOBI data — presented and compared in step 3 of the Results

Table 2: Stability class (by E_{hull} and experimental/theoretical character).

Class	n	E_{hull} range (eV/atom)
stable ($E_{\text{hull}} = 0$)	160	0.0000
experimental metastable	183	0.0000347 – 2.1080
theoretical metastable	242	0.0000843 – 3.1688
unknown*	12	—
Total	597	

* 2 COD references with no E_{hull} (S_4N_2 , S_4N_4), 6 validation structures without metadata (see Table 1), 1 further COD reference (ZnSn/NiAs), and 3 compounds inherited from the original campaign, labeled "metastable" before the `theoretical` flag existed.

section (§6.3), not here: classification is not an end in itself, it exists solely to condition the viability-descriptor correlation tested in that same section.

6 Results

Logic of this section

Four steps, in this order: (1) validate this project’s methodology by reproducing the 7 reference reactions chosen by Reitz & Dronskowski; (2) test this project’s central correlation — viability (a compound’s thermodynamic/kinetic stability) against descriptors derived from LOBSTER data (ICOHP, antibonding ICOHP, ICOBI, antibonding ICOBI); (3) classify compounds and elements into metallic/ionic/covalent/mixed directly from that same ICOHP/ICOBI data, and compare that classification against `classify()` — a tool in service of the next step, not an end in itself; (4) determine whether step 2’s correlation is strengthened or weakened once conditioned on step 3’s bond type.

6.1 Step 1 — Methodological validation: the 7 Reitz & Dronskowski reactions

Full detail: `analysis/REPORT_delta_icohp_viability.md`, `analysis/test_delta_icohp_viability.py`.

Per-bond-type ICOHP values for 7 reactions with independently known ΔICOHP (transcribed independently of any calculation in this project) are run end to end through `balance.py/delta.py/classify.py`:

Table 3: Validating the endobondic/exobondic classification on 7 reactions with independently known ΔICOHP .

Reaction	Reference (kJ/mol)	Computed (kJ/mol)	Label	
			Ref.	Computed
$\text{Pb}(\text{N}_3)_2 \rightarrow \text{Pb} + 3 \text{N}_2$	+1345	+1344.3	endo	endo
$\text{S}_4\text{N}_2 \rightarrow \frac{1}{2} \text{S}_8 + \text{N}_2$	+258	+258.5	endo	endo
$\text{S}_4\text{N}_4 \rightarrow \frac{1}{2} \text{S}_8 + 2 \text{N}_2$	+399	+397.3	endo	endo
$\text{ZnSn} \rightarrow \text{Zn} + \text{Sn}$	−337	−336.6	exo	exo
$\text{CaO}[\text{sphalerite}] \rightarrow \text{CaO}[\text{rocksalt}]$	−79	−78.9	exo	exo
$\text{CaN} \rightarrow \frac{1}{3} \text{Ca}_3\text{N}_2 + \frac{1}{6} \text{N}_2$	−205	−204.8	exo	exo
$\text{Mn}_2\text{O}_7 \rightarrow 2 \text{MnO}_2 + \frac{3}{2} \text{O}_2$	−186	−187.7	exo	exo

The 7 cases reproduce the reference ΔICOHP to within a few kJ/mol (source per-bond value rounding) and every `BondingLabel` matches. The Mn_2O_7 case is a useful guardrail on its own:

exobondic under this static sign criterion, yet Mn_2O_7 is a real, observed compound that simply decomposes slowly (gradual O_2 loss) rather than through abrupt bond breaking — `classify.py` structurally attaches a warning to *every* UNSTABLE_NONEXISTENT verdict for exactly this reason, rather than special-casing Mn_2O_7 .

This validation only compares the arithmetic of the manuscript’s own published ICOHP values, independent of any local crystal structure — Appendix K documents, for the 16 species involved, which MP/COD structure was used in this dataset and whether it could be verified to match the manuscript’s description (3 real corrections identified: N_2 , the $\text{Pb}(\text{N}_3)_2$ polymorph, ZnSn ’s competing tin phase). For N_2 and ZnSn , a DFT+LOBSTER calculation of this project’s own (not just the manuscript’s arithmetic) was additionally compared directly against the manuscript’s numbers (Table S18): N_2 reproduces the manuscript’s ICOHP to within 0.7%; ZnSn reproduces the same sign (bonding-favorable decomposition) but with $\sim 2.9\times$ larger magnitude — a gap judged not significant at this stage, since a bulk intermetallic’s ICOHP sum is far more sensitive to basis/pseudopotential/cutoff choices than a simple gas dimer. An independent campaign reproducing the manuscript’s full DFT methodology (PBEsol+D3, Table S1’s own k-point sets) across all 16 species, rather than just these two, is under way (§9).

Correlation alone (Pearson/Spearman) is not the right test for "do our numbers match the reference values" — it is satisfied by any monotonic relationship, even with a large systematic offset or scale factor. **Lin’s concordance correlation coefficient** (CCC) specifically tests agreement with the identity line (computed = reference), the claim actually being tested here:

Table 4: Computed-vs-reference concordance on the 7 validation reactions.

Statistic	Value
n	7
Lin’s CCC	0.999998
Pearson r	0.999999 ($p = 3.5 \times 10^{-15}$)
Sign agreement	7/7
Mean residual	0.76 kJ/mol
Max residual	1.70 kJ/mol
RMSE	0.98 kJ/mol

A CCC this close to 1 means the two methods agree essentially to the reporting precision of the reference values themselves (rounded to 2–3 decimals), not merely that they move together — the implementation is validated, not just spot-checked. It is this implementation, validated here independently of any of this project’s own calculations, that is applied to the full dataset in the next step.

6.2 Step 2 — Viability versus LOBSTER descriptors

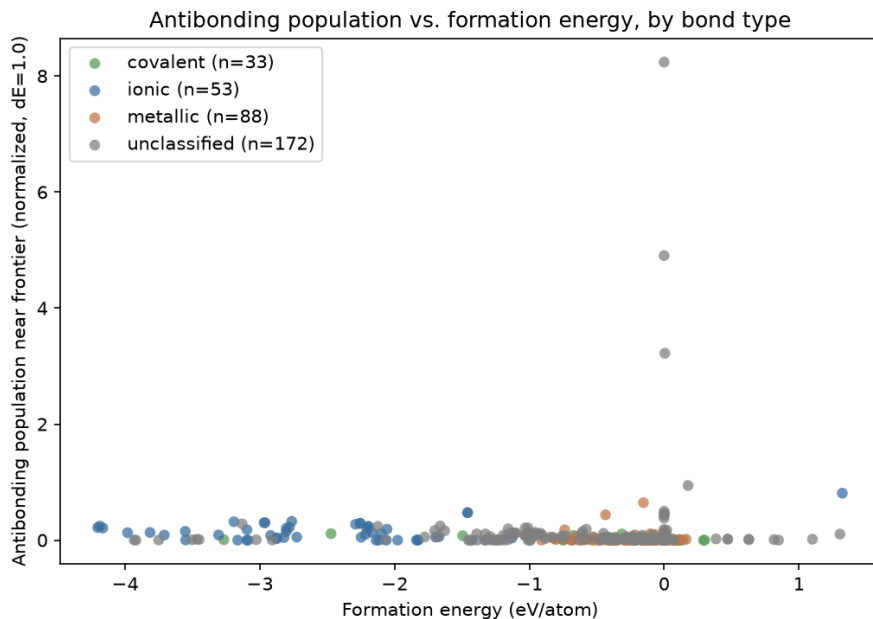
Classic ICOHP aggregates (baseline). Before any constructed descriptor, the raw ICOHP aggregates (sum, minimum per compound) are tested against E_{hull} as-is, on the original 186-compound campaign’s subset:

Table 5: Spearman correlations, classic ICOHP aggregates (sum, minimum) against E_{hull} , 186-compound subset.

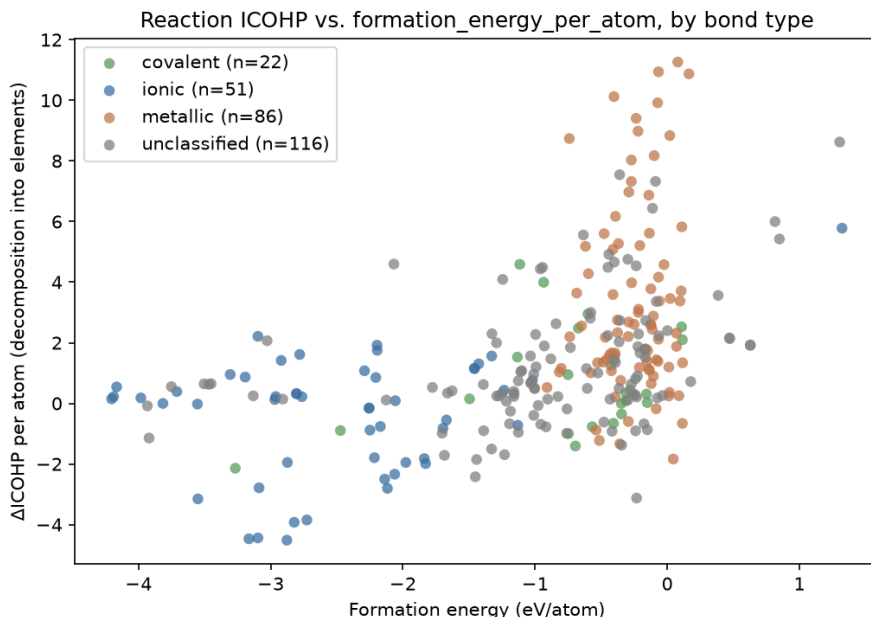
Group	Metric	n	ρ	p
all	ICOHP sum	186	0.032	0.670
all	ICOHP min	186	0.093	0.209
metallic	ICOHP sum	88	0.223	0.037
metallic	ICOHP min	88	0.198	0.065

These four rows correlate little or not at all with E_{hull} , except in the metallic subgroup — a weak baseline the descriptors below need to beat.

Antibonding population near the Fermi level (ICOHP). A distinct question from the fully-integrated ICOHP (§2.4): not *how much* bonding a compound has, but *how* COHP is distributed in energy — the occupied antibonding fraction just below the Fermi level/VBM, by analogy with Peierls/Jahn-Teller-type electronic instabilities. Result, against `formation_energy_per_atom`: $\rho = -0.155$, $p = 1.8 \times 10^{-4}$ ($n = 579$) — weaker than at smaller scale ($\rho = -0.282$, $n = 346$) but still real and significant; the sign remains the *opposite* of the naive Peierls/Jahn-Teller reading (§6.4 for the bond-type-stratified reading, where this descriptor’s headline stratified result no longer survives at the current scale).



Reaction ICOHP. An ICOHP analog of `formation_energy_per_atom`: is a compound’s total ICOHP "cheaper" than that of the same atoms as reference elements? Result: $\rho = 0.295$, $p \approx 0$ ($n = 510$) — a real, highly significant correlation given the sample size, with an intuitive sign (less bonding than in elemental form $\Rightarrow$ less stable formation). On a 288-compound subset shared with every other descriptor in the project, the antibonding population above slightly edges out reaction ICOHP on both targets — the latter remains a strong descriptor, but no longer unambiguously *the* strongest. Case 2 (direct same-formula polymorph comparison, 74 groups): 47.3% agreement between the most-bonded and the most-stable member, against a 42.4% chance baseline (exact Poisson-binomial test, $P = 0.22$) — no signal.



$\Delta(\text{ICOHP})$, applied to the full dataset, pooled. 517 case-1 reactions, every element and compound with a computed decomposition-into-elements reaction, pooled into a single population (grouping by selection provenance would be exactly the kind of between-group confound this project’s methodology is meant to catch):

Table 6: Spearman correlations, $\Delta\text{ICOHP}/\text{atom}$ (case 1, 517 reactions) against two continuous stability targets.

Target	n	ρ	p
formation_energy_per_atom	510	-0.2948	1.1×10^{-11}
energy_above_hull_eV_per_atom	514	-0.0564	0.202

(Sign: `reaction_analysis`’s products–reactants convention, opposite to `reaction_icohp.py`’s own column above — same magnitude, flipped sign.) Both correlations are weaker than at smaller scale: the formation-energy result was $\rho = -0.50$ before the marginal-formation-energy and max-hull-distance batches were folded in, and the hull-distance result moved from borderline ($p = 0.068$) to clearly non-significant ($p = 0.20$) — consistent with deliberately adding compounds with marginal or extreme thermodynamics, which dilute a rank correlation partly carried by the original population’s narrower stability range.

Sign-based discrimination (endobondic/exobondic). Experimental viability indicator (`theoretical=False` means experimentally realized):

Table 7: Endobondic/exobondic label against experimental/theoretical status.

	endobondic	exobondic
experimental ($n = 267$)	79	188
theoretical only ($n = 244$)	46	198

Fisher’s exact test: $p = 0.0054$, odds ratio = 1.81 — **significant** (endobondic is $\sim 1.8\times$ more frequent among experimentally realized compounds, the expected direction). Mann–Whitney on the continuous $\Delta\text{ICOHP}/\text{atom}$, same split: experimental median -0.868 eV, theoretical-only median -1.454 eV (more exobondic), $p = 0.0109$ — same direction, also significant. This is

exactly the result the marginal-formation-energy and max-hull-distance batches were built to test: the pooled 281-reaction population did not have enough near-stability-boundary compounds to reveal this signal ($p = 0.10$ at that scale).

Reading trap, explicitly flagged by `classify.py` itself: Table 7 shows 188 *experimental* (i.e. genuinely existing) compounds labeled *exobondic*. This is **not** a contradiction — exobondic only implies "not viable" once combined with `delta_energy` (`classify_viability()`): a compound is only `UNSTABLE_NONEXISTENT` if decomposition into elements is *both* exobondic *and* thermodynamically favorable ($\Delta E < 0$), which is rare (decomposing a real compound into pure elements is almost never favorable). Verified directly on the same population as Table 7:

Table 8: Computed `ViabilityLabel` (`classify_viability()`) on the same population as Table 7 ($n = 518$).

	experimental ($n = 267$)	theoretical ($n = 244$)
<code>STABLE_ON_HULL</code>	233	131
<code>UNSTABLE_NONEXISTENT</code>	26	95
<code>METASTABLE_VIABLE</code>	8	17

7 additional reactions (COD-sourced or otherwise `mp_id`-less, so no `formation_energy_per_atom`) excluded, insufficient data.

Collapsing `STABLE_ON_HULL+METASTABLE_VIABLE` ("viable") against `UNSTABLE_NONEXISTENT`: Fisher’s exact test $p = 2.9 \times 10^{-15}$, odds ratio **6.0** ($\chi^2 = 59.0$, $p = 1.6 \times 10^{-14}$) — **a much larger effect than the sign-only test** (Table 7, $p = 0.0054$, odds ratio 1.81). An experimentally realized compound is roughly $6\times$ less likely to be predicted `UNSTABLE_NONEXISTENT` than a theoretical-only one. **Of the 188 experimental exobondic compounds, only 26 are predicted `UNSTABLE_NONEXISTENT`** — the other 162 are `STABLE_ON_HULL`: their decomposition into elements, though exobondic, is not thermodynamically favorable. Reading Table 7 in isolation as "exobondic = not viable" is therefore misleading — it is Table 8, combining sign with thermodynamics, that actually answers the viability question the **theoretical** flag is a proxy for, and it is by far the strongest result of this step.

Table 9: Median $\Delta\text{ICOHP}/\text{atom}$ and % endobondic by stability class ($n = 177$).

Class	n	median $\Delta\text{ICOHP}/\text{atom}$ (eV)	% endobondic
experimental metastable	59	−1.038	28.8%
stable	59	−1.391	22.0%
theoretical metastable	59	−1.601	13.6%

Kruskal–Wallis across the three groups: $H = 2.56$, $p = 0.278$ — **not significant** (population unchanged at $n = 177$: the extension/marginal-formation-energy/max-hull batches carry a **theoretical** flag but not yet a **family** label, so this particular test has not benefited from the dataset’s growth), but the ordering remains exactly what physics predicts: experimentally metastable compounds have the highest endobondic fraction; never-synthesized theoretical metastable compounds have the lowest.

Step reading. The 7 reference reactions reproduce almost exactly (step 1) — the implementation is correct. Extended to the current full dataset (517 reactions, up from 281), the coarse sign-based discrimination is now significant, and the full `ViabilityLabel` machinery — sign combined with thermodynamics — is far more strongly so. The continuous correlations against `formation_energy_per_atom` and `energy_above_hull` weakened over the same period — the expected price of adding chemically diverse, thermodynamically extreme compounds to

what was a narrower population; a different kind of test from the categorical results above, so the two are not in tension. Whether the `ViabilityLabel` result is itself confounded by bond type (step 4), or by the fact that max-hull’s theoretical polymorphs were deliberately selected far above the hull — a `formation_energy_per_atom`-adjacent selection effect, not necessarily an independent confirmation of the bonding-kinetics mechanism — remains to be checked.

Still no SISSO at this stage: detailed reports, `analysis/REPORT_antibonding.md`, `analysis/REPORT_reaction_icohp.md`.

6.3 Step 3 — Bond-type classification from ICOHP/ICOBI data

A tool in service of the next step, not an end in itself: this classification exists solely to test, in step 4, whether step 2’s viability-descriptor correlation depends on a compound’s bond type. Two independent classifications (§2.2): the `classify()` function (elemental composition + `is_metal`) and a classification built directly from each compound’s ICOHP/ICOBI data (`is_metal` for metallic character; otherwise the ICOBI of the **dominant first-coordination-shell** species pair — not a mean over every neighbor LOBSTER reports within a wider cutoff, which strongly dilutes the signal for dense covalent solids such as diamond — compared against a threshold calibrated at the midpoint of the median ICOBI of compounds `classify()` already labels ionic/covalent, i.e. 0.5315, to separate ionic from covalent among non-metallic compounds). A fourth label, **mixed** (Zintl-type), identifies compounds where a *homoatomic* pair above the covalent threshold (e.g. N–N in an azide, P–P in a polyphosphide, S–S in a polysulfide) coexists with a markedly weaker *heteroatomic* pair linking a distinct cation — the structural signature of Zintl–Klemm chemistry, where a single ionic/covalent label would be misleading.

Table 10: Bond type: `classify()` vs. the ICOHP/ICOBI classification (concordance on the subset `classify()` labels: $276/311 = 88.7\%$).

<code>classify()</code> \ ICOHP/ICOBI	metallic	ionic	covalent	mixed	unclassified
metallic	193	0	0	0	0
ionic	0	48	2	3	0
covalent	0	14	35	16	0
unclassified	179	36	19	43	0
Total	372	98	56	62	0

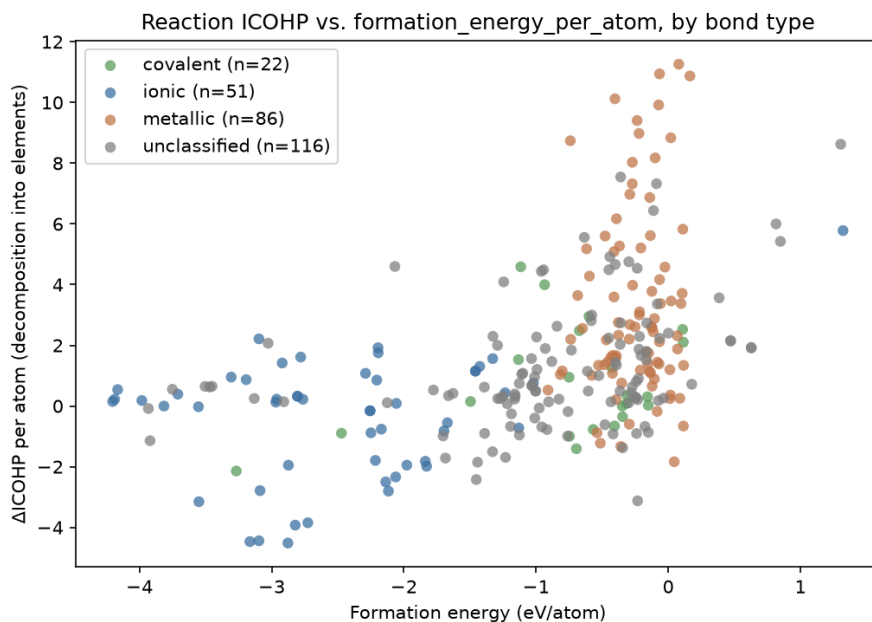
Of the 277 compounds `classify()` leaves "unclassified" (essentially alkali/alkaline-earth $\times$ {N, P, S} pairs outside `IONIC_ANIONS`, and metals bonded to an anion-like element that `classify()` deliberately excludes from its "metallic" bucket), the ICOHP/ICOBI classification resolves **55** into ionic (36) or covalent (19), and identifies **43** as mixed (Zintl-type); the remaining 179 are metallic, already covered by `is_metal`. Where both methods render a verdict (metallic/ionic/covalent), agreement is strong (88.7%) but not total: 16 ionic/covalent disagreements, and 19 compounds `classify()` labels ionic or covalent but that the ICOBI classification identifies as mixed (their dominant homoatomic bond coexists with a markedly weaker heteroatomic link) — worth checking case by case before using either classification as a stratification variable in step 4. It is the ICOHP/ICOBI classification (broader coverage, built on the same data as step 2’s descriptors) that is used below, not `classify()`.

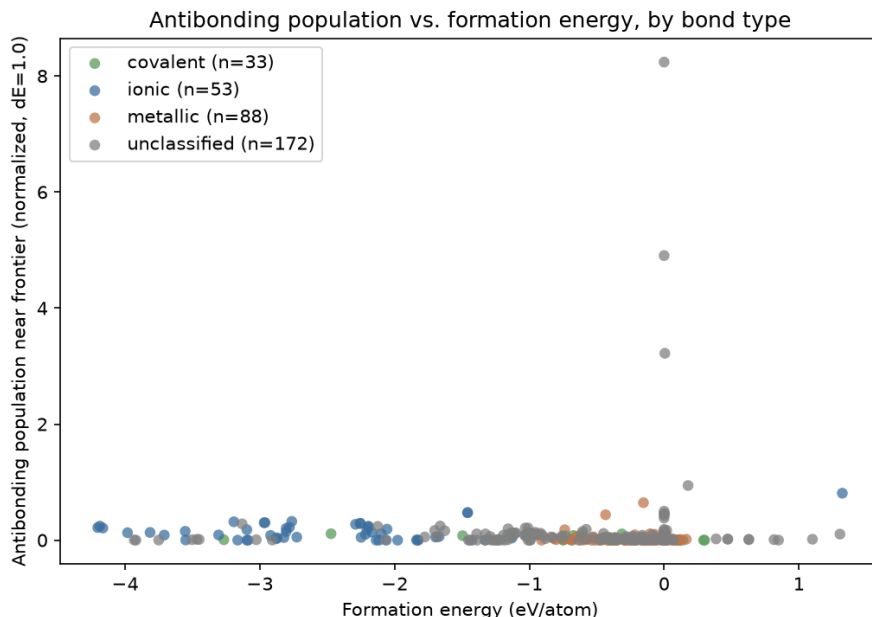
6.4 Step 4 — Does the viability-descriptor correlation depend on bond type?

Antibonding population (ICOHP). The project’s previously most robust result is **retracted**: `bond_type=covalent`, which survived clearly at $n = 33$ ($\rho = -0.551$, $p = 0.0009$), no longer survives once the sample grew to $n = 54$ ($\rho = -0.046$, $p = 0.74$) — a failure mode already seen elsewhere in this project (a subgroup result that does not replicate once the sample

grows), including this same descriptor’s own earlier `bond_type=ionic` subgroup. **New at this scale:** `bond_type=mixed` (Zintl) is now the best-surviving stratum ($\rho = -0.538$, $p = 6.6 \times 10^{-6}$, $n = 62$) — **strengthened** relative to step 2’s pooled correlation — a category that barely existed when the covalent result was first reported. `is_metal=False` remains the most consistently surviving stratum across this project’s successive scales ($\rho = -0.333$, $p = 6.7 \times 10^{-7}$, $n = 212$) — **strengthened**, not weakened, by the dataset’s growth. `is_metal=True` has also converged onto `bond_type=metallic` (identical $n = 367$, $\rho = 0.056$, not significant), the same phenomenon documented below for reaction ICOHP.

Reaction ICOHP. `bond_type` stratification (now four categories, including `mixed`/Zintl) no longer uniformly dilutes step 2’s pooled correlation ($\rho = 0.295$): `ionic` ($\rho = 0.360$, $p = 0.0004$) and `mixed` ($\rho = -0.525$, opposite sign, $p \approx 0$) both survive clearly — **strengthened** in magnitude — `covalent` too but flips sign ($\rho = -0.354$, $p = 0.023$); only `metallic` (the majority class, 315/510 rows) stays non-significant — **weakened** to the point of disappearing in that stratum. `is_metal` stratification — the one that used to survive — has now converged **onto the same population** as `bond_type=metallic` (identical n , ρ , p in every row) and so inherits its non-significant result: it is `bond_type`, not `is_metal`, that carries information at the current dataset composition. Against `energy_above_hull`, `bond_type=mixed` again survives on its own ($\rho = -0.307$, $p = 0.016$).





Synthesis. In both cases (antibonding and reaction ICOHP), this step’s answer is *yes, but not uniformly*: bond type does not merely weaken or strengthen a single global correlation, it changes the *sign* of the relationship depending on the stratum (ionic positive, covalent/mixed negative for reaction ICOHP) — the useful signal is now conditional on bond type. The antibonding population’s covalent-result retraction is itself a direct illustration of why this step exists: a stratified number is only trustworthy once re-checked as the dataset grows, not accepted once and for all. A pooled feature search (SISSO) would therefore be actively misleading here; any future search should be run separately per `bond_type`, not on the pooled population.

7 Failures Found and Fixed

Per the instruction ("every job failure must be logged with its probable cause"), 5 of the first 178 SLURM-array-submitted jobs failed on the first pass, all diagnosed and fixed at the source (the fixes, applied to the shared SLURM/INCAR templates, carry over to the whole dataset):

- **Al₁₂W, BW, WO₂, TcW, TiW** (all tungsten-containing): the local mirror’s `W_sv` POT-CAR is missing its `LEXCH` field entirely (confirmed by direct inspection — present in every other variant checked, including plain `W`), which VASP reads as an incompatible exchange-correlation functional and refuses to run. Fixed by replacing the `W_sv` override with plain `W` (valid PBE, verified).
- **Al₁₂W** failed again after that fix, this time with a VASP SIGSEGV — a known issue (`ulimit -s unlimited` before `vasp_std`), present in `submit_array.sh` but forgotten in `prepare_vasp_lobster.py`’s per-compound SLURM template. Fixed; recovered on retry.

Final success rate across all 597 structures computed to date: **597/597** for VASP; **596/597** for LOBSTER (1 trivial-cell reference element with no usable LOBSTER step, §8).

8 Compute Cost

597 structures computed in total: 69 single-element references, 522 multi-element compounds, and 6 further validation structures without compound metadata but with their own VASP+LOBSTER run (§5).

Every VASP job, then LOBSTER job, runs on the **Yargla** HPC cluster, within a single SLURM allocation of one node (`-nodes=1`) and **16 cores**: VASP under MPI parallelism (`-ntasks=16 -cpus-per-task=1, srun`), then LOBSTER under OpenMP parallelism on the same 16 cores (`OMP_NUM_THREADS=16`, single process, run sequentially after VASP within the same job). The original SLURM array (`submit_array.sh`, main campaign) is capped at 8 concurrent tasks; the batches added since (extension/marginal-formation-energy/ max-hull/widen) were submitted individually (`sbatch` per directory), without that cap of their own but sharing the cluster with other, unrelated workloads.

Table 11: VASP and LOBSTER compute time (597 structures).

	VASP	LOBSTER
n	597	596*
Mean (s)	188	578
Median (s)	72	157
Max (s)	5715 (1.6 h)	22555 (6.3 h)
Sequential total	31.2 h	95.8 h

* 1 of 597 structures had no usable LOBSTER step (trivial-cell reference element).

Combined sequential total (VASP + LOBSTER, 597 structures): **127.0 h**. The main campaign (capped at 8 concurrent tasks) gave an order-of-magnitude elapsed time of roughly one-eighth of the sequential total; that factor no longer applies uniformly across the whole dataset — the individually-submitted batches added since have no cap of their own but shared the cluster queue with other projects, making a single elapsed-time estimate for the full dataset not meaningful.

9 Current Limits and Next Steps

- **One compound excluded for LOBSTER quality:** `extension_Cu4Au_mp-1225761` was dropped from every analysis (but not from the raw dataset – its VASP+LOBSTER calculation is kept for reference) — `lobsterout` reports 343/343 k-points failed orthonormalization (100%, not a borderline case), and `bandOverlaps.lobster` shows a maximum deviation of 11.4 (this project’s own established red-flag threshold is 0.1). Direct consequence: physically implausible ICOHP values (third- neighbor Cu–Cu contacts at 4.25–5.04 Å with ICOHP values of 7.6–11.6 eV) and a $\Delta\text{ICOHP}/\text{atom}$ of -380.5 eV, $\sim 27\times$ the next-worst case in the rest of the population. A systematic sweep of all 597 structures (`analysis/audit_lobster_quality.py`, criterion: 100% k-point failure *and* deviation > 0.5 *and* robust z-score > 10) confirms this is a unique case, not a wider problem — the second-worst case (**Ir4W**) is already $29\times$ less extreme and plausibly real chemistry (a dense, d-electron-rich intermetallic). Every statistic in this report covering the 518 case-1 reactions has been regenerated on the remaining 517 — the effect on any given number is negligible (3rd–4th decimal place), no conclusion or significance threshold is affected.
- **classify() coverage:** 277/588 compounds (47%) fall outside the ionic/covalent/metallic categories, mostly alkali/alkaline-earth $\times$ {N, P, S} bonds that `IONIC_ANIONS` does not recognize (§5) — purely informative here, but a real limitation for any mission that would want to stratify on it.
- **No true dynamical "unstable" class:** `theo_metastable` is a proxy (never synthesized), not a phonon-calculation result.

- **Per-subgroup sample size:** remains this project’s structural limit at every new stratification (`bond_type`, `is_metal`) — see §6.2 for a concrete example where an $n < 15$ subgroup did not replicate once the sample grew.
- **Why not SISSO yet:** reaction ICOHP’s signal (§6.2) is now conditional on bond type and opposite-signed across strata (ionic positive, covalent/mixed negative) — a pooled feature search would be misleading here; any future search should be run separately per `bond_type`.
- **family not yet extended to the newer batches:** the theoretical-based `ViabilityLabel`/Fisher discrimination (§6.2) fully benefits from the dataset’s growth, but the finer three-way split (`exp_metastable/exp_stable/theo_metastable`) stays at $n = 177$ for lack of a `family` label on the extension/marginal-formation-energy/max-hull batches — the direct next step to test whether that finer split also turns significant.
- **The `ViabilityLabel` result (§6.2, $p = 2.9 \times 10^{-15}$) has not yet been checked for confounds:** neither stratified by `bond_type/is_metal`, nor distinguished from a selection effect (max-hull’s theoretical polymorphs were deliberately chosen far above the hull).

10 Conclusion

Reitz & Dronskowski’s endobondic/exobondic methodology (§6.2) reproduces faithfully across the 7 independently known reference reactions (Lin’s CCC 0.999998) and applies to this report’s full dataset (518 decomposition-into-elements reactions). On that population, sign-based Δ ICOHP discrimination is now significant (Fisher’s test, $p = 0.0054$), and the full `ViabilityLabel` machinery — sign combined with thermodynamics — is far more strongly so ($p = 2.9 \times 10^{-15}$, odds ratio 6.0): an experimentally realized compound is roughly $6\times$ less likely to be predicted `UNSTABLE_NONEXISTENT` than a theoretical-only one (§6.2). Reaction ICOHP otherwise remains a strong continuous correlate against formation energy ($\rho = 0.295$, §6.2), though antibonding population now slightly edges it out on the subset shared by both descriptors.

Two concrete directions follow. First, checking whether the `ViabilityLabel` result above survives stratification by `bond_type/is_metal` and is not simply carried by max-hull’s deliberate selection effect — the same verification discipline this project applies systematically to every new descriptor. Second, restricting the reaction Δ (ICOHP) to a window near the Fermi level rather than the fully integrated ICOHP, modeled on the antibonding population of §6.2: two variants of this construction are detailed in Appendix J (antibonding part only, and the full signed ICOHP restricted to the window) — the first survives every stratification tested, a result no other descriptor in this report reaches.

Source code: <https://github.com/John-Akapulco/viability>

Data: `analysis/percolation_vs_formation_energy.csv`

Detailed reports: `analysis/REPORT_*.md`

Appendix (SI)

A Compound and Element List

Complete list of the 392 compounds and elements with usable ICOBI data (of 394 total; 2 COD-sourced compounds lack `is_metal`/ICOBI, §5), organized by section according to the independent ICOHP/ICOBI classification of §5 (`is_metal` for metallic character; otherwise the ICOBI

of the **dominant first-coordination-shell** species pair — not a mean over every neighbor LOBSTER reports within the wider percolation-graph cutoff — compared against a threshold of 0.5052 to separate ionic from covalent; a **mixed** (Zintl-type) category groups compounds where a strongly covalent homoatomic pair coexists with a markedly weaker heteroatomic pair, e.g. the azide ion N_3^- ionically bound to its cation) — no “bond type” column; each section instead states its own observed ICOBI range. ICOBI and ICOHP are per-bond means (§2.4); the ICOHP/ICOBI antibonding populations are the normalized values at $\Delta E = 1.0$ eV (§6.2, §J).

A.1 Metallic

Table S1: Metallic compounds and elements (`is_metal=True`, observed ICOBI 0.0000 to 2.9506) — formula, mp-id, space group, dominant (first-shell) bond ICOBI, dominant bond ICOHP, normalized ICOHP and ICOBI antibonding populations ($\Delta E = 1.0$ eV).

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
Ag*	mp-124	Fm-3m	0.1545	-1.2496	0.0000	0.0000
AgN*	mp-1091417	Cmce	0.4624	-3.4181	0.0088	0.0030
AgN	mp-1009766	F-43m	0.4854	-3.5762	0.0277	0.0173
AgN	mp-1424734	Ama2	0.8238	-5.9446	0.0107	0.0025
AgN	mp-1428312	Pm-3m	0.2177	-1.4991	0.0083	0.0041
Al*	mp-134	Fm-3m	0.2270	-1.4193	0.0000	0.0000
Al ₁₂ W*	mp-11227	Im-3	0.5027	-2.7492	0.0000	0.0000
Al ₃ Os ₂ *	mp-16521	I4/mmm	0.3022	-1.6476	0.0037	0.0000
Al ₄ Si	mp-1228120	R-3m	0.2614	-2.0375	0.0087	0.0030
AlNi	—	—	0.2676	-1.9848	0.0016	0.0023
AlSb	mp-1023918	Cmcm	0.2647	-1.2802	0.0000	0.0000
As*	mp-158	Cmce	0.8848	-3.9683	0.0103	0.0000
AsPd ₂ *	mp-1080831	P-62m	0.3410	-2.4185	0.0077	0.0157
AsRh*	mp-22079	Pnma	0.3323	-1.8639	0.0449	0.0158
AsRh ₂ *	mp-1102364	Pnma	0.2438	-1.3778	0.1158	0.0189
Au*	mp-81	Fm-3m	0.1962	-1.3381	0.0000	0.0000
B ₃ Ir ₂	mp-1228647	C2/m	0.4473	-3.3600	0.0047	0.0011
BPd ₅	mp-1228665	I4/mmm	0.4754	-3.9924	0.0973	0.0181
BW*	mp-7832	I4_1/amd	0.4277	-3.9556	0.0000	0.0000
Ba*	mp-122	Im-3m	0.0002	-0.0122	8.2400	2.9573
BaAu ₂ *	mp-30363	P6/mmm	0.6152	-3.0712	0.0144	0.0034
BaF ₃	mp-1183303	I4/mmm	0.0350	-0.5277	0.2805	0.2537
BaN ₂ *	mp-1102371	Cc	0.1605	-0.4471	0.1039	0.0214
BaSn ₅ *	mp-571353	P6/mmm	0.2221	-0.8507	0.0456	0.0042
Be*	mp-87	P6_3/mmc	0.2033	-1.8975	0.0000	0.0000
Be ₁₂ Pd*	mp-12666	I4/mmm	0.2078	-1.9466	0.0000	0.0000
Be ₂ Cu*	mp-2031	Fd-3m	0.3616	-3.0863	0.0000	0.0001
Be ₂ Nb ₃ *	mp-11272	P4/mbm	0.4200	-3.5436	0.0000	0.0000
Be ₂ V*	mp-11281	P6_3/mmc	0.3658	-2.7859	0.0000	0.0000
BeAu ₃	mp-983539	Pm-3m	0.2230	-1.9129	0.0000	0.0030
BeCu	—	—	0.1984	-1.8359	0.0000	0.0000
BeF ₂	mp-975644	P6/mmm	0.1548	-1.6978	0.0863	0.0000
Bi ₂ O ₃ *	mp-558953	P2_1/c	0.9517	-6.7444	0.0328	0.0038
Bi ₂ O ₃	mp-1383505	P1	0.9506	-7.1985	0.0609	0.0000
C*	mp-48	P6_3/mmc	1.2195	-11.5387	0.0000	0.0000
Ca*	mp-21	Im-3m	0.0025	-0.0304	3.2206	3.3021
Ca ₃ N ₂	mp-1013524	Pm-3m	0.1107	-0.3623	0.0090	0.0000
CaAl ₄ *	mp-1749	I4/mmm	0.4595	-1.8701	0.0086	0.0002

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
CaGa ₂ *	mp-992	P6/mmm	0.9791	-3.8887	0.0219	0.0000
CaGa ₄ *	mp-2461	C2/m	0.4641	-2.1536	0.0084	0.0001
CaGe ₂	mp-643542	C2/m	0.7379	-3.1645	0.0337	0.0038
CaN*	mp-1058549	Fm-3m	0.1201	-0.6934	0.0450	0.0656
CaO	mp-1181903	Fd-3m	0.0725	-0.5516	0.1865	0.0756
CaSn*	mp-2450	Cmcm	0.7044	-2.1965	0.0629	0.0276
Cd*	mp-94	P6 ₃ /mmc	0.2423	-1.6119	0.0000	0.0000
CdPd*	mp-1696	P4/mmm	0.2022	-1.5904	0.0000	0.0004
Co*	mp-102	Fm-3m	0.2680	-1.9150	0.0189	0.0373
CoB*	mp-20857	Pnma	0.4568	-4.5928	0.0099	0.0000
CoSe ₂	mp-1390196	Fd-3m	0.7048	-4.3267	0.0348	0.0079
Cr*	mp-90	Im-3m	0.2487	-0.6245	0.0115	0.0049
Cr ₂ C	mp-1226378	P-3m1	0.5270	-3.0803	0.0084	0.0000
CrMo	mp-1226200	Cmmm	0.3600	-1.3679	0.0081	0.0026
Cs*	mp-1	Im-3m	0.0556	-0.1230	0.1696	0.0693
CsO ₂ *	mp-1441	I4/mmm	1.0942	-11.4627	0.1143	0.1436
CsO ₂	mp-1096936	Cmmm	1.1298	-11.0556	0.1197	0.0872
Cu*	mp-30	Fm-3m	0.0456	-0.2488	0.1641	0.0946
Cu ₃ Ge*	mp-19724	Pmmn	0.1369	-0.8340	0.0963	0.0410
Fe*	mp-13	Im-3m	0.1417	-0.5370	0.0922	0.0794
FeS*	mp-505531	P4/nmm	0.4967	-2.5883	0.0736	0.0024
FeTe ₂ *	mp-1102002	Pa-3	0.4279	-1.7861	0.0401	0.0000
Ga*	mp-142	Cmce	0.4308	-2.6378	0.0000	0.0000
Ge*	mp-32	Fd-3m	0.8956	-4.6450	0.0015	0.0000
Ge ₂ Os*	mp-10032	C2/m	0.4042	-2.1066	0.0069	0.0003
Ge ₄ Rh*	mp-1104286	P3 ₁ 21	0.3355	-1.8036	0.0034	0.0006
Hf*	mp-103	P6 ₃ /mmc	0.3121	-1.1612	0.0000	0.0000
HfMo	mp-1224314	Cmmm	0.3368	-1.2722	0.0000	0.0000
HfTc ₂ *	mp-1095669	P6 ₃ /mmc	0.3775	-1.4429	0.0098	0.0012
Hg*	mp-10861	P6/mmm	0.1622	-0.9652	0.0000	0.0000
Hg ₄ Pt*	mp-936	Im-3m	0.5477	-3.5588	0.0000	0.0005
IN ₂	mp-1097011	I4/mcm	2.9506	-23.1289	0.0177	0.0275
In*	mp-1055994	I4/mmm	0.2177	-1.1562	0.0000	0.0000
In ₃ Ir*	mp-630976	P4 ₂ /mnm	0.5910	-3.4701	0.0000	0.0000
InSe ₂	mp-1232036	P4/nmm	0.2771	-1.4780	0.0349	0.0038
Ir*	mp-101	Fm-3m	0.3469	-2.4130	0.0317	0.0274
K*	mp-10157	Fm-3m	0.0430	-0.0808	0.0923	0.0499
K ₂ O	mp-1223784	P4/mmm	0.1375	-0.3763	0.0831	0.0000
K ₅ As ₄	mp-684799	P-1	0.9859	-3.9306	0.0452	0.0189
KN ₂ *	mp-1071868	Cmc2 ₁	1.0490	-7.9889	0.0202	0.0000
KN ₂	mp-1180757	Cmcm	1.0210	-7.4152	0.0411	0.0000
KN ₃	mp-636056	I4/mcm	2.8711	-22.8747	0.0090	0.0107
KP*	mp-1058315	R-3m	0.1254	-0.4533	0.0190	0.0442
KP	mp-1057787	Immm	0.1070	-0.4621	0.0170	0.0218
KSn ₃	mp-1185151	P6 ₃ /mmc	0.4006	-1.5431	0.0179	0.0027
Li*	mp-1018134	R-3m	0.0886	-0.6931	0.0000	0.0000
Li ₁₃ Sn ₅ *	mp-30769	P-3m1	0.6069	-2.7881	0.0000	0.0000
Li ₃ Ag	mp-976408	I4/mmm	0.1084	-1.0444	0.0000	0.0000
Li ₃ Hg*	mp-1646	Fm-3m	0.1488	-1.2194	0.0000	0.0000
LiB*	mp-1001835	P6 ₃ /mmc	1.1571	-9.9815	0.0000	0.0000
LiTl ₃	mp-1185253	I4/mmm	0.2370	-1.2989	0.0000	0.0000
Mg*	mp-153	P6 ₃ /mmc	0.0000	0.0000	—	—
Mg ₁₅ B	mp-1023515	P-6m2	0.0280	-0.1152	0.9423	0.5909
Mg ₂ Ni*	mp-2137	P6 ₂ 22	0.0818	-0.2478	0.6457	0.3329
MgPd ₃ *	mp-7753	Pm-3m	0.2542	-1.7837	0.0264	0.0304
MgSn	mp-1094669	Pm-3m	0.2238	-0.7050	0.1000	0.0334

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
MgSn	mp-1094801	P4/mmm	0.3134	-1.2760	0.1099	0.0281
Mn*	mp-35	I-43m	—	-0.6082	0.0411	0.0047
MnAl ₁₂ *	mp-1104165	Im-3	0.2798	-1.6283	0.0000	0.0000
MnO ₂ *	mp-510408	P4 ₂ /mmn	0.4433	-3.0527	0.0548	0.0603
Mo*	mp-129	Im-3m	0.3113	-1.1059	0.0139	0.0005
Mo ₃ Se ₄ *	mp-21021	R-3	0.4924	-2.4124	0.0151	0.0004
N ₂	mp-1059834	Imma	1.3533	-15.9343	0.0159	0.0010
Na*	mp-10172	P6 ₃ /mmc	0.0470	-0.0839	0.1168	0.0901
Na ₂ Cl	mp-1077015	Cmmm	0.0377	-0.1951	0.0901	0.0029
Na ₂ Cl	mp-990084	R-3m	0.0479	-0.3527	0.0618	0.0000
Na ₃ Cl	mp-1064484	P4/mmm	0.0297	-0.2254	0.0859	0.0249
Nb*	mp-75	Im-3m	0.3268	-1.1439	0.0000	0.0000
Nb ₃ Ir*	mp-1458	Pm-3n	0.4144	-2.0103	0.0000	0.0002
NbGa ₃ *	mp-1973	I4/mmm	0.3104	-1.7262	0.0000	0.0000
NbO*	mp-2692	Fm-3m	0.2933	-2.2812	0.0292	0.0023
NbO	mp-634965	P4/mmm	0.9184	-5.5839	0.0000	0.0000
NbP*	mp-9830	I4 ₁ /amd	0.3299	-1.7803	0.0001	0.0000
NbP	mp-1220317	Cmmm	0.3238	-1.7805	0.0084	0.0004
NbRh ₂	mp-1220414	Pm	0.3585	-1.4459	0.0440	0.0023
NbS ₂	mp-774873	P6 ₃ /mmc	0.6390	-3.1946	0.0049	0.0000
Ni*	mp-23	Fm-3m	0.0947	-0.4204	0.3799	0.2037
NiB*	mp-14019	Cmcm	0.6996	-6.1857	0.0198	0.0093
NiHg ₄ *	mp-570835	Im-3m	0.3125	-2.2179	0.0000	0.0004
NiS*	mp-1547	R3m	0.2630	-1.5895	0.1140	0.0342
Os*	mp-49	P6 ₃ /mmc	0.2567	-1.1579	0.0538	0.0135
OsC*	mp-7142	P-6m2	0.4688	-3.0941	0.0402	0.0068
OsC	mp-1009537	Pm-3m	0.3353	-2.0943	0.0446	0.0000
OsC	mp-1009539	Fm-3m	0.4618	-3.1760	0.0828	0.0058
OsC	mp-999308	P6 ₃ /mmc	0.4521	-3.1583	0.0525	0.0226
Pb*	mp-20483	Fm-3m	0.1943	-0.8519	0.0276	0.0000
PbAu ₂ *	mp-19871	Fd-3m	0.3048	-2.3586	0.0000	0.0000
Pd*	mp-2	Fm-3m	0.1631	-1.2162	0.1264	0.0877
Pt*	mp-126	Fm-3m	0.2367	-1.8089	0.0927	0.0819
Rb*	mp-70	Im-3m	0.0590	-0.1180	0.1307	0.0377
Rb ₂ Hg ₇ *	mp-31474	P-3m1	0.1920	-0.9499	0.0000	0.0000
RbP*	mp-1179688	C2/m	0.1031	-0.4288	0.0138	0.0467
RbP	mp-1058499	Immm	0.1277	-0.4304	0.0099	0.0324
Re*	mp-8	P6 ₃ /mmc	0.3162	-1.3803	0.0205	0.0030
Re ₃ Ge ₇ *	mp-29141	Cmcm	0.4553	-2.2942	0.0022	0.0002
ReC*	mp-1079494	P6 ₃ /mmc	0.4741	-3.3056	0.0211	0.0022
ReC	mp-1009731	Pm-3m	0.3588	-2.2826	0.0149	0.0003
Rh*	mp-74	Fm-3m	0.1620	-0.7081	0.1852	0.0924
Ru*	mp-33	P6 ₃ /mmc	0.2375	-0.9478	0.0875	0.0211
RuC*	mp-10030	P-6m2	0.4770	-2.9147	0.0418	0.0078
RuC	mp-1009787	Fm-3m	0.4752	-2.9734	0.0862	0.0067
SN	mp-1078816	P2 ₁ /c	0.2823	-1.7903	0.0359	0.0006
SN	mp-1173453	P2 ₁	1.5195	-10.9244	0.0458	0.0000
SN	mp-684642	P2 ₁	1.3504	-13.4315	0.0333	0.0038
Sb*	mp-104	R-3m	0.4398	-1.7202	0.0425	0.0063
Sc*	mp-67	P6 ₃ /mmc	0.2110	-0.6596	0.0000	0.0000
Sc ₅ Ge ₃ *	mp-17190	P6 ₃ /mcm	0.3162	-1.5145	0.0000	0.0000
ScAg ₂ *	mp-1018128	I4/mmm	0.2014	-1.5516	0.0000	0.0000
ScTc ₃	mp-867262	Fm-3m	0.3043	-1.0739	0.0080	0.0010
ScTe*	mp-10026	P6 ₃ /mmc	0.3365	-1.5673	0.0000	0.0000
SiAu ₃	mp-1186998	Fm-3m	0.1974	-1.4121	0.0156	0.0000
SiO ₂	mp-556044	P-31c	0.7437	-6.4399	0.0117	0.0000

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
SiO ₂	mp-638049	F4_132	0.6594	-6.0556	0.0488	0.0000
SiRh [*]	mp-2287	P2_1/c	0.2801	-1.6004	0.0243	0.0009
Sn [*]	mp-623511	I4/mmm	0.1989	-0.8803	0.0235	0.0006
Sn ₇ Ir ₅ [*]	mp-20466	I4mm	0.5375	-3.2800	0.0000	0.0000
Sn ₇ Pd ₅	mp-1219006	C2	0.3765	-2.7926	0.0000	0.0004
SnPt [*]	mp-19856	P6_3/mmc	0.4612	-3.2535	0.0000	0.0000
Sr [*]	mp-139	P6_3/mmc	0.0009	-0.0171	4.9050	5.3659
Sr ₂ As [*]	mp-15697	I4/mmm	0.0787	-0.2746	0.2097	0.0267
Sr ₃ Sn	mp-1187136	Pm-3m	0.0501	-0.2374	0.4377	0.0368
Sr ₅ Sn ₃ [*]	mp-17720	I4/mcm	0.8182	-2.1022	0.1796	0.0000
SrAl ₂ [*]	mp-1638	Fd-3m	0.4263	-1.6649	0.0004	0.0000
SrN [*]	mp-1058586	Fm-3m	0.1230	-0.6070	0.0462	0.0849
SrSi [*]	mp-2661	Cmcm	0.8077	-2.8883	0.0564	0.0097
SrTl ₃	mp-1206636	Pm-3m	0.2857	-1.2068	0.0201	0.0007
Ta [*]	mp-50	Im-3m	0.3226	-1.2794	0.0000	0.0000
Ta ₄ C ₃	mp-1218000	R3m	0.5265	-3.5440	0.0000	0.0000
Ta ₅ Ge ₃ [*]	mp-17593	I4/mcm	0.3556	-2.6739	0.0000	0.0000
Ta ₇ Co ₆ [*]	mp-1104548	R-3m	0.4891	-3.0211	0.0018	0.0000
TaN [*]	mp-1497	P6/mmm	0.4241	-2.9806	0.0009	0.0000
TaN	mp-1009831	Pm-3m	0.3185	-2.1377	0.0002	0.0002
TaN	mp-1009833	F-43m	0.6451	-4.4476	0.0000	0.0006
TaP [*]	mp-1067587	I4_1md	0.4762	-2.7480	0.0000	0.0000
TaP	mp-1187244	P-6m2	0.4792	-2.7509	0.0000	0.0000
TaRe	mp-1217894	Cmmm	0.3311	-1.5114	0.0026	0.0003
TaSb ₂ [*]	mp-11697	C2/m	0.4899	-2.0334	0.0002	0.0000
Tc [*]	mp-113	P6_3/mmc	0.3130	-1.2061	0.0240	0.0006
TcW	mp-1217337	Cmmm	0.6035	-3.6836	0.0000	0.0002
Te ₂ Au [*]	mp-567525	C2/m	0.5036	-3.0627	0.0497	0.0327
Te ₂ Ir [*]	mp-1551	C2/m	0.7621	-4.4689	0.0230	0.0046
Te ₂ Pd [*]	mp-782	P-3m1	0.5813	-3.5060	0.0291	0.0200
Ti [*]	mp-46	P6_3/mmc	0.3010	-0.7536	0.0000	0.0000
Ti ₃ Ge [*]	mp-1189044	I-4	0.3243	-1.6051	0.0000	0.0000
Ti ₃ In [*]	mp-866046	P6_3/mmc	0.3128	-1.1764	0.0000	0.0000
Ti ₃ Pt [*]	mp-2339	Pm-3n	0.3632	-1.7698	0.0000	0.0000
TiBe ₁₂ [*]	mp-1104067	I4/mmm	0.2186	-1.9904	0.0000	0.0000
TiMo ₂	mp-1216675	Fmmm	0.3395	-1.1740	0.0000	0.0000
TiNi [*]	mp-1048	P2_1/m	0.2818	-0.8685	0.0059	0.0000
TiNi	mp-1067248	Cmcm	0.2798	-0.8586	0.0060	0.0000
TiO [*]	mp-1071163	P-62m	0.3594	-2.4547	0.0015	0.0000
TiSi ₂ [*]	mp-1077503	Cmcm	0.4599	-3.0700	0.0000	0.0000
TiSi ₂	mp-570991	Immm	0.2659	-1.2939	0.0000	0.0000
TiW	mp-1216621	Cmmm	0.6568	-3.6207	0.0000	0.0000
Tl [*]	mp-82	P6_3/mmc	0.1905	-1.0463	0.0000	0.0000
Tl ₃ Ga	mp-1187539	Fm-3m	0.2090	-1.0914	0.0000	0.0000
Tl ₄ Sn	mp-1216560	R-3m	0.2039	-1.3837	0.0010	0.0008
TlC [*]	mp-567465	Fm-3m	0.3776	-2.4265	0.0000	0.0010
TlC	mp-1058887	P4/mmm	0.4837	-3.1370	0.0000	0.0003
TlN [*]	mp-1017982	P6_3mc	0.6066	-3.6824	0.0023	0.0000
TlN	mp-1002222	Fm-3m	0.3446	-2.1516	0.0029	0.0000
TlN	mp-1007778	F-43m	0.6105	-3.6815	0.0017	0.0000
V [*]	mp-146	Im-3m	0.3178	-0.8777	0.0000	0.0000
VCo ₃	mp-1095057	P6_3/mmc	0.3730	-2.5141	0.0012	0.0023
VN	mp-1017532	P6_3/mmc	0.4293	-2.6954	0.0000	0.0000
VNi ₂ [*]	mp-11531	Immm	0.4769	-1.4029	0.0918	0.0008
W [*]	mp-91	Im-3m	0.4372	-2.5600	0.0000	0.0001
WC [*]	mp-13136	Fm-3m	0.5508	-4.2240	0.0012	0.0009

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
WC	mp-1008630	Pm-3m	0.4127	-2.9246	0.0000	0.0010
WC	mp-1008635	F-43m	0.8710	-6.2790	0.0000	0.0000
WO ₂ *	mp-1524394	P2 ₁ /c	0.6713	-5.7408	0.0039	0.0000
Y*	mp-112	P6 ₃ /mmc	0.2374	-0.8682	0.0000	0.0000
Y ₃ Co	mp-1189329	P2 ₁	0.4718	-2.6045	0.0000	0.0000
YAg ₃	mp-1187811	Fm-3m	0.1780	-1.2749	0.0000	0.0000
Zn*	mp-79	P6 ₃ /mmc	0.2296	-1.8892	0.0000	0.0000
Zn	mp-2647117	Im-3m	0.1819	-1.3641	0.0000	0.0000
ZnAg*	mp-1912	Pm-3m	0.2035	-1.8265	0.0000	0.0000
ZnPb ₃	mp-1187949	Fm-3m	0.2392	-1.3227	0.0004	0.0004
Zr*	mp-131	P6 ₃ /mmc	0.3223	-1.3170	0.0000	0.0000
Zr ₂ Au*	mp-2348	I4/mmm	0.3095	-2.0203	0.0000	0.0000
Zr ₃ Be	mp-1188016	I4/mmm	0.3655	-1.7390	0.0000	0.0000
Zr ₃ Ge	mp-1188039	P6 ₃ /mmc	0.3460	-1.7196	0.0000	0.0000
Zr ₅ Pb ₃ *	mp-681992	P6 ₃ /mcm	0.3301	-1.9739	0.0000	0.0000
ZrBr*	mp-1065846	C2/m	0.3528	-2.0185	0.0000	0.0001
ZrGa ₃ *	mp-30685	I4/mmm	0.2933	-1.8532	0.0000	0.0000
ZrMo ₂ *	mp-2049	Fd-3m	0.4671	-1.6863	0.0002	0.0000
ZrRh*	mp-2808	Pm-3m	0.3062	-1.4732	0.0094	0.0024
ZrTi	mp-1215236	P4/mmm	0.3777	-1.6592	0.0000	0.0000

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §2.4).

A.2 Ionic

Table S2: Ionic compounds (non-metallic, ICOBI < 0.5052, observed range 0.0444 to 0.4976) — same columns.

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
AgO*	mp-1079720	P2 ₁ /c	0.2604	-2.0408	0.0297	0.0024
B ₆ Pb	mp-1096825	Pm-3m	0.2353	-1.9376	0.0000	0.0000
BaCl ₂ *	mp-568662	Fm-3m	0.0749	-0.5008	0.2998	0.0066
BaCl ₂ *	mp-23199	Pnma	0.0637	-0.4308	0.3027	0.0019
BaF ₃ *	mp-1080821	P2 ₁ /m	0.2279	-0.7883	0.3169	0.0437
BaH ₂ *	mp-23715	Pnma	0.0615	-0.2536	0.1495	0.0000
BaO*	mp-1008500	P6 ₃ /mmc	0.0788	-0.6380	0.1468	0.0016
BaO	mp-1950989	P6 ₃ /mmc	0.1070	-0.4367	0.2225	0.0000
Be ₃ N ₂ *	mp-6977	P6 ₃ /mmc	0.4361	-4.2580	0.0000	0.0000
Be ₃ N ₂ *	mp-18337	Ia-3	0.4759	-4.7015	0.0000	0.0000
Be ₃ N ₂	mp-1017550	P-3m1	0.3952	-3.8753	0.0000	0.0000
BeCl ₂	mp-1172909	I-43m	0.3697	-0.9454	0.8095	0.1761
Ca ₃ N ₂ *	mp-1047	R-3c	0.1781	-0.6015	0.1048	0.0000
Ca ₃ N ₂ *	mp-844	Ia-3	0.1820	-0.6211	0.0837	0.0000
CaCl ₂ *	mp-23214	Pnnm	0.1054	-0.7540	0.2048	0.0008
CaCl ₂	mp-1120739	Fm-3m	0.0814	-0.5708	0.3236	0.0068
CaF ₂ *	mp-10464	Pnma	0.0565	-0.5566	0.2424	0.0000
CaF ₂	mp-554355	P4/mmm	0.0591	-0.5440	0.1286	0.0000
CaO*	mp-1079707	Cmce	0.1008	-0.5793	0.1805	0.0000
CaO*	mp-2605	Fm-3m	0.1143	-0.7028	0.0897	0.0000
CdI ₂ *	mp-567259	P-3m1	0.4859	-2.4996	0.0508	0.0000
Cs ₂ S*	mp-1077814	P6 ₃ /mmc	0.0804	-1.0011	0.0682	0.0000
Cs ₂ Se*	mp-1011696	Pnma	0.0811	-0.6189	0.0861	0.0000
CsCl*	mp-573697	Fm-3m	0.0595	-0.3453	0.0490	0.0000
CsF*	mp-8455	Pm-3m	0.1224	-1.5102	0.0523	0.0008

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
CsH*	mp-632319	Pm-3m	0.0935	-1.2305	0.0678	0.0002
GeS	mp-12910	Cmcm	0.4348	-2.2462	0.0516	0.0000
H ₂ O*	mp-2739191	C222_1	0.4372	-4.6426	0.0087	0.0000
HI*	mp-697025	C2/c	0.4915	-2.7099	0.0522	0.0000
HgF ₂ *	mp-8177	Fm-3m	0.2320	-1.8744	0.1660	0.0000
I*	mp-23153	Cmce	0.0444	-0.1504	0.4981	0.0900
InBr*	mp-22870	Cmcm	0.3010	-1.3292	0.0929	0.0024
InCl*	mp-571636	Cmc2_1	0.1698	-0.8368	0.0921	0.0006
K ₂ O*	mp-971	Fm-3m	0.1009	-0.8814	0.0059	0.0000
K ₂ S*	mp-559278	P6_3/mmc	0.1040	-0.4891	0.0071	0.0006
K ₂ S*	mp-1022	Fm-3m	0.1212	-0.4744	0.0000	0.0000
KCl*	mp-23289	Pm-3m	0.0510	-0.4078	0.1374	0.0000
KF ₃ *	mp-1544557	P2_12_12_1	0.3162	-0.9256	0.4749	0.3545
KF ₃	mp-2749823	P2_1	0.3119	-0.9244	0.4733	0.3514
KI*	mp-22898	Fm-3m	0.0893	-0.4549	0.0615	0.0000
Li ₂ O*	mp-1960	Fm-3m	0.2393	-2.0320	0.0000	0.0000
Li ₂ O	mp-755894	Pnma	0.2225	-2.0369	0.0000	0.0000
Li ₂ S*	mp-1125	Pnma	0.2847	-2.0194	0.0000	0.0000
Li ₂ S	mp-557142	P6_3/mmc	0.3327	-2.1900	0.0000	0.0000
Li ₂ Se*	mp-2286	Fm-3m	0.3399	-2.1702	0.0000	0.0000
Li ₃ N*	mp-2341	P6_3/mmc	0.2587	-1.8972	0.0000	0.0000
LiBr	—	—	0.2790	-2.0381	0.0000	0.0000
LiCl*	mp-22905	Fm-3m	0.2558	-2.0554	0.0000	0.0000
LiCl	mp-1185319	P6_3mc	0.3879	-2.9688	0.0000	0.0000
LiF*	mp-1138	Fm-3m	0.1692	-1.9334	0.0001	0.0000
LiF	mp-1009009	Pm-3m	0.1237	-1.3411	0.0249	0.0000
MgCl ₂ *	mp-570259	P-3m1	0.1312	-0.9334	0.2905	0.0028
MgCl ₂ *	mp-23210	R-3m	0.1313	-0.9344	0.2883	0.0009
MgCl ₂	mp-569626	Ama2	0.1475	-1.0096	0.1898	0.0001
MgF ₂ *	mp-1072956	Pnnm	0.0901	-1.0192	0.1330	0.0000
MgF ₂	mp-1062842	Amm2	0.0808	-0.8465	0.1522	0.0000
MgF ₂	mp-560236	C2/m	0.1007	-1.0813	0.0878	0.0000
MgH ₂ *	mp-23711	Pbcn	0.1156	-0.5595	0.0617	0.0000
MgO*	mp-549706	P6_3mc	0.1572	-1.2025	0.0812	0.0000
MgO	mp-1009127	Pm-3m	0.0813	-0.6164	0.2735	0.0000
MgS*	mp-1315	Fm-3m	0.1236	-0.7426	0.2438	0.0000
MgS	mp-13032	F-43m	0.2006	-1.1035	0.1624	0.0000
Na ₂ O*	mp-2352	Fm-3m	0.1191	-0.6209	0.0000	0.0000
Na ₂ O	mp-1975297	C2/m	0.1297	-0.6320	0.0003	0.0000
NaCl	—	—	0.0859	-0.5953	0.1111	0.0000
NaS*	mp-409	P-62m	0.4976	-2.6241	0.1126	0.0000
NaS	mp-1180106	C2/m	0.2104	-0.9839	0.0000	0.0000
PbSe*	mp-22009	Fmm2	0.4169	-1.6445	0.0576	0.0000
PbSe*	mp-2201	Fm-3m	0.3457	-1.4608	0.0807	0.0039
Rb ₂ O*	mp-1394	Fm-3m	0.0867	-1.2998	0.0368	0.0000
Rb ₂ O	mp-2023482	P-3m1	0.1302	-0.4067	0.0636	0.0000
RbCl*	mp-23299	Pm-3m	0.0478	-0.6732	0.1019	0.0000
RbF*	mp-2064	Pm-3m	0.0879	-1.3700	0.0447	0.0036
RbF	mp-11718	Fm-3m	0.0460	-0.4536	0.0398	0.0000
RbI*	mp-22903	Fm-3m	0.0840	-0.4162	0.0534	0.0013
RbS	mp-1179684	C2/m	0.1226	-0.5861	0.0850	0.0092
Sc ₂ O ₃	mp-755313	R-3c	0.4210	-2.7698	0.0026	0.0000
SiO ₂	mp-10064	Fm-3m	0.3974	-3.7939	0.0240	0.0000
SnBr ₂	mp-978985	P4_2/mnm	0.3595	-1.6332	0.1289	0.0000
SrF ₂ *	mp-1102240	Pnma	0.0589	-0.5401	0.2090	0.0000
SrF ₂	mp-1102911	Pnma	0.0505	-0.4858	0.2166	0.0000

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
SrS*	mp-10627	Pm-3m	0.0762	-0.4641	0.2376	0.0000
Te*	mp-19	P3_121	0.3172	-1.1233	0.1197	0.0127
Te ₂ Ru*	mp-267	Pnnm	0.4723	-2.2556	0.0486	0.0017
TlCl*	mp-571079	Cmcm	0.1717	-0.8409	0.1079	0.0000
Y ₂ O ₃ *	mp-558573	C2/m	0.4214	-2.9843	0.0000	0.0000

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §2.4).

A.3 Covalent

Table S3: Covalent compounds (non-metallic, ICOBI ≥ 0.5052 , observed range 0.5087 to 1.9179) — same columns.

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
As ₂ S ₃ *	mp-1189572	P-1	0.9136	-4.7416	0.0538	0.0000
AsF ₃ *	mp-28027	Pna2_1	0.6951	-6.3410	0.1114	0.0000
B*	mp-160	R-3m	0.5203	-4.7960	0.0000	0.0000
BI ₃ *	mp-23189	P6_3/m	1.2727	-6.8061	0.0232	0.0000
Be ₃ P ₂ *	mp-567841	Ia-3	0.5637	-3.8282	0.0000	0.0000
Be ₃ P ₂	mp-1084771	Pn-3m	0.5501	-3.7626	0.0000	0.0002
BeCl ₂ *	mp-23267	Ibam	0.7165	-5.6352	0.0021	0.0000
BeCl ₂	mp-1190080	I-43m	0.7528	-5.9613	0.0026	0.0000
BeF ₂ *	mp-15951	P3_121	0.5269	-6.8585	0.0035	0.0000
BeO*	mp-7599	P4_2/mnm	0.5148	-5.7846	0.0000	0.0000
BeO	mp-1778	F-43m	0.5232	-5.8507	0.0000	0.0000
Bi ₂ O ₃	mp-714859	P1	1.1110	-10.1326	0.0646	0.0001
Br*	mp-23154	Cmce	0.7827	-3.5362	0.4257	0.0097
C	mp-1080826	C2/m	0.9446	-9.5768	0.0000	0.0000
C	mp-1190171	Pnma	0.9445	-9.5798	0.0000	0.0000
C*	mp-66	Fd-3m	0.9497	-9.6602	0.0000	0.0000
C*	mp-47	P6_3/mmc	0.9482	-9.7832	0.0000	0.0000
C	—	—	1.2173	-11.6891	0.0000	0.0000
C ₃ N ₄ *	mp-571653	P-43m	0.9531	-10.4956	0.0150	0.0000
C ₃ N ₄	mp-1182484	P6_3/m	1.9179	-15.3586	0.0000	0.0000
C ₃ N ₄	mp-2852	I-43d	0.9575	-10.9099	0.0106	0.0000
C ₃ N ₄	mp-563	Fd-3m	0.7047	-7.1477	0.0086	0.0000
Cl ₂ *	mp-22848	Cmce	0.9254	-5.5728	0.4555	0.0003
ClF ₃ *	mp-556767	Pnma	0.6390	-5.2173	0.1347	0.0000
F ₂ *	mp-561203	C2/c	0.9731	-6.9536	0.4569	0.0000
Ga ₂ Se ₃	mp-1224809	Imm2	0.8324	-4.7105	0.0085	0.0000
GeSe ₂ *	mp-10074	I-42d	0.9011	-4.9682	0.0171	0.0000
H ₂ *	mp-730101	P2_12_12_1	0.9954	-5.1222	0.0000	0.0000
H ₃ N*	mp-643432	P2_12_12_1	0.8816	-8.0639	0.0000	0.0000
HCl	mp-684609	Imm2	0.9480	-7.3985	0.0478	0.0001
IF ₇ *	mp-27988	Aea2	0.5305	-5.5662	0.0568	0.0000
IrCl ₃	mp-729507	C2	0.7112	-4.9995	0.0957	0.0000
Mn ₂ O ₇ *	mp-28338	P2_1/c	1.3609	-6.7339	0.0257	0.0000
MnN	mp-1009130	F-43m	0.6682	-3.9792	0.0247	0.0062
MoSe ₂ *	mp-1634	P6_3/mmc	0.6411	-2.8467	0.0035	0.0000
Nb ₂ O ₅ *	mp-1595	C2/m	0.7548	-4.9296	0.0075	0.0000
O ₂ *	mp-1524462	C2/m	1.4816	-17.7956	0.1271	0.2186
P*	mp-568348	P2/c	0.9202	-5.0760	0.0209	0.0000
PS*	mp-612	C2/c	0.9840	-5.8992	0.0301	0.0000
PtCl ₂	mp-1219748	C2/m	0.7183	-5.3013	0.0756	0.0000

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
S*	mp-77	Fddd	0.9692	-5.9246	0.0880	0.0000
SN*	mp-236	P2_1/c	1.1480	-8.7413	0.0440	0.0000
Se*	mp-14	P3_121	0.9053	-4.1496	0.0828	0.0000
Si	—	—	0.9243	-4.5418	0.0000	0.0000
SiO ₂ *	mp-546794	I-42d	0.7466	-7.9383	0.0071	0.0000
SiO ₂ *	mp-9258	Pa-3	0.5130	-5.5969	0.0264	0.0000
SiS ₂ *	mp-1095294	P2_1/c	0.9089	-5.7585	0.0044	0.0000
TeO ₂ *	mp-8377	P2_12_12_1	0.6215	-4.7653	0.0772	0.0000
TiO ₂ *	mp-1439	Pbcn	0.5838	-3.5796	0.0077	0.0000
TiO ₂ *	mp-9173	Pnma	0.5897	-3.5580	0.0123	0.0000
TiO ₂ *	mp-430	P2_1/c	0.5087	-3.0819	0.0145	0.0000
ZrO ₂	mp-775909	P4_2/mnm	0.5720	-4.1446	0.0017	0.0000

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §2.4).

A.4 Mixed (Zintl-type)

Table S4: Mixed-bonding (Zintl-type) compounds (a strongly covalent homoatomic pair coexists with a markedly weaker heteroatomic pair, e.g. the azide N_3^- ion ionically bound to its cation; observed range 0.8522 to 2.7322) — same columns.

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
Ba ₅ P ₄ *	mp-17566	Pnma	0.9040	-3.4927	0.2104	0.0000
BaS ₃ *	mp-556296	P2_12_12	0.9201	-5.2850	0.1084	0.0000
Bi ₂ O ₃	mp-674644	P1	0.9332	-7.5456	0.0131	0.0000
CdO ₂ *	mp-2310	Pa-3	0.9738	-7.1680	0.0540	0.0000
Cs ₂ As ₃ *	mp-15557	Fmmm	1.0431	-4.4248	0.0386	0.0000
CsN ₃ *	mp-510557	I4/mcm	1.8888	-17.7624	0.0000	0.0000
CsN ₃	mp-1225892	P4/mmm	1.8983	-17.5725	0.0000	0.0000
CsP ₇ *	mp-1201790	Pbcm	0.9343	-4.9186	0.0062	0.0000
CsP ₇	mp-1213206	Pca2_1	0.9336	-4.9318	0.0064	0.0000
KN ₃ *	mp-827	I4/mcm	1.8889	-18.4973	0.0000	0.0000
LiP ₅ *	mp-2412	Pna2_1	0.9037	-4.9376	0.0026	0.0000
LiP ₅	mp-32760	Pna2_1	0.8771	-4.6811	0.0064	0.0000
MgP ₄ *	mp-384	P2_1/c	0.9196	-5.0216	0.0273	0.0000
NaN ₃ *	mp-1066400	R3m	1.8853	-18.8113	0.0000	0.0000
NaN ₃ *	mp-22003	R-3m	1.8872	-19.1626	0.0002	0.0000
NaN ₃	mp-1065265	C2/m	2.7322	-23.2128	0.0006	0.0215
NaS*	mp-2400	P6_3/mmc	0.9510	-4.7236	0.1279	0.0000
PI ₂ *	mp-29443	P-1	0.9586	-4.8005	0.1081	0.0000
PbN ₆ *	mp-667338	Pnma	1.8687	-19.2227	0.0007	0.0000
RbN ₃ *	mp-581833	P4/mmm	1.8969	-18.3403	0.0000	0.0000
RbN ₃	mp-1219567	Amm2	1.3756	-9.8572	0.0216	0.0000
RbS*	mp-558071	Immm	0.9920	-5.0554	0.1964	0.0019
Sr ₃ As ₄	mp-678007	P1	0.8522	-3.0556	0.1161	0.0000
Sr ₃ P ₄ *	mp-14288	Fdd2	0.8910	-4.1353	0.1224	0.0000
Sr ₃ P ₄	mp-682953	P1	0.8889	-4.0705	0.1250	0.0000
SrO ₂ *	mp-2697	I4/mmm	0.9835	-7.0573	0.1841	0.0000
SrO ₂ *	mp-726705	Pnma	0.9821	-6.8700	0.2389	0.0000
SrO ₂	mp-2763179	Pnma	0.9805	-6.9738	0.2327	0.0000

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §2.4).

A.5 Unclassified

Table S5: Unclassified compounds (no ICOBI data available) — same columns, ICOBI/antibonding shown as “–” where missing.

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
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Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §2.4).

B VASP Calculation Parameters

Fixed INCAR parameters, identical across the whole dataset (static, LOBSTER-compatible run) — only NBANDS (appendix D) and the k-point grid vary by compound.

Parameter	Value	Description
PREC	Accurate	Overall precision (charge grid, projectors)
ENCUT	600.0	Plane-wave cutoff energy (eV)
EDIFF	1e-06	Electronic convergence criterion (eV)
IBRION	-1	Ionic dynamics type (−1 = static, no relaxation)
NSW	0	Number of ionic steps (0 = static run on the MP structure)
ISMear	0	Smearing type (0 = Gaussian)
SIGMA	0.05	Smearing width (eV)
LASPH	True	Non-spherical corrections to the PAW potential
ISYM	-1	Symmetry (−1 = disabled, required by LOBSTER)
LWAVE	True	Write WAVECAR (required by LOBSTER)
LCHARG	True	Write CHGCAR
NPAR	4	Band-level parallelization
KPAR	2	K-point-level parallelization

C PAW-PBE Potentials

One PAW-PBE potential per element present in the dataset (pymatgen’s `MPRelaxSet` library — the same one Materials Project used to relax the input structures), with one documented exception for tungsten (§7): the recommended `W_pv` variant is absent from the local mirror, and `W_sv` turned out to be corrupted (missing `LEXCH` field) — plain `W` is used instead, verified valid.

Table S6: PAW-PBE potentials used, one per element present in the dataset (pymatgen’s `MPRelaxSet` library, with one documented exception: `W` substituted for `W_pv`/`W_sv`, see §7).

Element	PAW-PBE potential
Ag	Ag (02Apr2005)
Al	Al (04Jan2001)
As	As (22Sep2009)
Au	Au (04Oct2007)

Element	PAW-PBE potential
B	B (06Sep2000)
Ba	Ba_sv (06Sep2000)
Be	Be_sv (06Sep2000)
Bi	Bi (08Apr2002)
Br	Br (06Sep2000)
C	C (08Apr2002)
Ca	Ca_sv (06Sep2000)
Cd	Cd (06Sep2000)
Cl	Cl (06Sep2000)
Co	Co (02Aug2007)
Cr	Cr_pv (02Aug2007)
Cs	Cs_sv (25Jan2019)
Cu	Cu_pv (06Sep2000)
F	F (08Apr2002)
Fe	Fe_pv (02Aug2007)
Ga	Ga_d (06Jul2010)
Ge	Ge_d (03Jul2007)
H	H (15Jun2001)
Hf	Hf_pv (06Sep2000)
Hg	Hg (06Sep2000)
I	I (08Apr2002)
In	In_d (06Sep2000)
Ir	Ir (06Sep2000)
K	K_sv (06Sep2000)
Li	Li_sv (10Sep2004)
Mg	Mg_pv (13Apr2007)
Mn	Mn_pv (02Aug2007)
Mo	Mo_pv (04Feb2005)
N	N (08Apr2002)
Na	Na_pv (19Sep2006)
Nb	Nb_pv (08Apr2002)
Ni	Ni_pv (06Sep2000)
O	O (08Apr2002)
Os	Os_pv (20Jan2003)
P	P (06Sep2000)
Pb	Pb_d (06Sep2000)
Pd	Pd (04Jan2005)
Pt	Pt (04Feb2005)
Rb	Rb_sv (06Sep2000)
Re	Re_pv (06Sep2000)
Rh	Rh_pv (25Jan2005)
Ru	Ru_pv (28Jan2005)
S	S (06Sep2000)
Sb	Sb (06Sep2000)
Sc	Sc_sv (07Sep2000)
Se	Se (06Sep2000)
Si	Si (05Jan2001)
Sn	Sn_d (06Sep2000)
Sr	Sr_sv (07Sep2000)
Ta	Ta_pv (07Sep2000)
Tc	Tc_pv (04Feb2005)
Te	Te (08Apr2002)
Ti	Ti_pv (07Sep2000)
Tl	Tl_d (06Sep2000)
V	V_pv (07Sep2000)
W	W (08Apr2002)

Element	PAW-PBE potential
Y	Y_sv (25May2007)
Zn	Zn (06Sep2000)
Zr	Zr_sv (04Jan2005)

D k-point Grid and NBANDS per Compound

k-point grid auto-generated by `pymatgen.io.vasp.Kpoints.automatic_density_by_vol` (density `kppvol=100`, Γ -centered or Monkhorst–Pack mesh depending on cell symmetry) and `NBANDS`, derived per compound from the number of local orbitals in the LOBSTER basis (`Lobsterin.write_INCAR`) — a strictly sufficient value for the local-basis projection, not a fixed guess.

Table S7: k-point grid (`pymatgen` automatic density, `kppvol=100`) and number of bands (derived from the LOBSTER basis) per compound.

Formula	mp-id	k-point grid	NBANDS
Experimental, stable (hull) ($n = 63$)			
As ₂ S ₃	mp-1189572	5 3 2 (G)	80
AsF ₃	mp-28027	5 4 4 (G)	64
BI ₃	mp-23189	4 4 4 (G)	32
BW	mp-7832	9 9 3 (G)	52
BaAu ₂	mp-30363	6 6 7 (G)	23
BaCl ₂	mp-568662	6 6 6 (G)	13
BaH ₂	mp-23715	6 4 3 (G)	28
Be ₂ Cu	mp-2031	7 7 7 (G)	38
Be ₂ Nb ₃	mp-11272	8 4 4 (M)	74
Be ₂ V	mp-11281	7 7 4 (G)	76
CaAl ₄	mp-1749	7 7 4 (G)	21
CaSn	mp-2450	6 6 4 (M)	28
CdPd	mp-1696	7 6 6 (G)	36
CoB	mp-20857	9 7 5 (G)	52
Cs ₂ As ₃	mp-15557	4 4 3 (G)	44
FeS	mp-505531	8 8 5 (G)	26
Ge ₂ Os	mp-10032	9 6 4 (G)	54
HfTc ₂	mp-1095669	5 5 3 (G)	108
Hg ₄ Pt	mp-936	5 5 5 (G)	45
HgF ₂	mp-8177	8 8 8 (G)	17
InBr	mp-22870	6 6 4 (M)	26
KI	mp-22898	6 6 6 (G)	9
KN ₃	mp-827	5 5 5 (G)	34
Li ₂ Se	mp-2286	7 7 7 (G)	14
LiB	mp-1001835	7 7 9 (G)	18
Mg ₂ Ni	mp-2137	5 5 2 (G)	102
MgP ₄	mp-384	5 5 3 (G)	40
MnAl ₁₂	mp-1104165	4 4 4 (M)	57
MoSe ₂	mp-1634	9 9 2 (G)	34
NaS	mp-2400	6 6 3 (G)	32
Na	mp-10172	8 8 4 (G)	8
Nb ₃ Ir	mp-1458	5 5 5 (G)	72
NbGa ₃	mp-1973	8 8 6 (M)	36
PI ₂	mp-29443	6 4 3 (G)	24
PbSe	mp-2201	7 7 7 (G)	13

Table S7 (continued)

Formula	mp-id	k-point grid	NBANDS
RbI	mp-22903	6 6 6 (G)	9
Sc ₅ Ge ₃	mp-17190	3 3 5 (G)	154
ScAg ₂	mp-1018128	8 8 5 (G)	28
ScTe	mp-10026	7 7 4 (G)	28
SnPt	mp-19856	7 7 5 (G)	36
SrO ₂	mp-2697	8 8 7 (G)	13
SrSi	mp-2661	7 6 4 (G)	18
Ta ₅ Ge ₃	mp-17593	4 4 4 (M)	144
TaP	mp-1067587	9 9 4 (G)	26
TaSb ₂	mp-11697	8 5 4 (G)	34
Te ₂ Pd	mp-782	7 7 5 (G)	17
Te ₂ Ru	mp-267	7 5 4 (G)	34
TeO ₂	mp-8377	6 5 3 (G)	48
Ti ₃ In	mp-866046	5 5 6 (G)	72
Ti ₃ Pt	mp-2339	5 5 5 (G)	72
TiBe ₁₂	mp-1104067	7 5 5 (G)	69
TiNi	mp-1048	10 7 6 (G)	36
TiO	mp-1071163	10 6 6 (G)	39
TiSi ₂	mp-1077503	8 8 4 (M)	34
TlCl	mp-571079	6 6 4 (M)	26
VNi ₂	mp-11531	12 8 7 (G)	27
WO ₂	mp-1524394	6 5 5 (G)	68
Zr ₂ Au	mp-2348	9 9 4 (G)	29
Zr ₅ Pb ₃	mp-681992	3 3 5 (G)	154
ZrMo ₂	mp-2049	6 6 6 (G)	56
Si	mp-149	8 8 8 (G)	100
NaCl	mp-22862	8 8 8 (G)	100
AlNi	mp-1487	10 10 10 (M)	100
Experimental, metastable ($n = 63$)			
CdI ₂	mp-567259	7 7 4 (G)	17
MgCl ₂	mp-570259	8 8 5 (G)	12
C	mp-169	12 12 8 (M)	100
ZrBr	mp-1065846	8 8 3 (G)	28
Li ₃ Hg	mp-1646	7 7 7 (G)	24
Ta ₇ Co ₆	mp-1104548	6 6 3 (G)	117
CaGa ₄	mp-2461	7 7 5 (G)	41
InCl	mp-571636	6 6 4 (M)	26
NiB	mp-14019	10 10 7 (G)	26
MgH ₂	mp-23711	6 5 5 (G)	24
Sn ₇ Ir ₅	mp-20466	4 4 4 (M)	108
NaN ₃	mp-1066400	7 7 7 (G)	16
ClF ₃	mp-556767	6 4 3 (G)	64
SiS ₂	mp-1095294	5 4 3 (G)	48
Sr ₅ Sn ₃	mp-17720	3 3 3 (G)	104
Te ₂ Ir	mp-1551	7 5 2 (G)	51
Li ₁₃ Sn ₅	mp-30769	6 6 1 (G)	110
H ₃ N	mp-643432	8 5 5 (G)	28
Cs ₂ Se	mp-1011696	5 3 2 (G)	56
In ₃ Ir	mp-630976	4 4 4 (M)	144
HI	mp-697025	3 3 3 (G)	40
Rb ₂ Hg ₇	mp-31474	4 4 4 (G)	73
PbAu ₂	mp-19871	5 5 5 (G)	54
BeCu	mp-2323	10 10 10 (M)	100
SiO ₂	mp-546794	6 6 6 (M)	24

Table S7 (continued)

Formula	mp-id	k-point grid	NBANDS
IF ₇	mp-27988	5 5 3 (G)	64
K	mp-10157	6 6 6 (G)	5
Al ₃ Os ₂	mp-16521	9 9 3 (G)	30
Nb ₂ O ₅	mp-1595	7 7 4 (G)	38
LiBr	mp-23259	8 8 8 (G)	100
ZnAg	mp-1912	9 9 9 (G)	18
MgPd ₃	mp-7753	7 7 7 (G)	31
AsPd ₂	mp-1080831	8 4 4 (G)	66
GeSe ₂	mp-10074	5 5 5 (G)	34
Re ₃ Ge ₇	mp-29141	9 6 1 (G)	180
Cu ₃ Ge	mp-19724	6 6 5 (G)	72
Ti ₃ Ge	mp-1189044	4 4 4 (M)	144
ZrGa ₃	mp-30685	7 7 5 (G)	37
SrAl ₂	mp-1638	5 5 5 (G)	26
NiS	mp-1547	6 6 6 (M)	39
Al ₁₂ W	mp-11227	4 4 4 (M)	57
CsH	mp-632319	7 7 7 (G)	6
CaGa ₂	mp-992	7 7 6 (G)	23
NiHg ₄	mp-570835	6 6 6 (M)	45
SiRh	mp-2287	6 6 5 (G)	52
BaSn ₅	mp-571353	5 5 4 (G)	50
PS	mp-612	4 4 3 (G)	64
AsRh	mp-22079	8 5 4 (G)	52
Te ₂ Au	mp-567525	7 7 5 (G)	17
Be ₃ N ₂	mp-6977	10 10 3 (G)	46
Y ₂ O ₃	mp-558573	8 4 3 (G)	96
AgO	mp-1079720	8 5 5 (G)	52
FeTe ₂	mp-1102002	4 4 4 (M)	68
H ₂ O	mp-2739191	7 7 4 (G)	24
CdO ₂	mp-2310	5 5 5 (G)	68
Mo ₃ Se ₄	mp-21021	4 4 4 (M)	86
PbSe	mp-22009	6 6 2 (G)	26
Be ₁₂ Pd	mp-12666	6 6 6 (M)	69
Sr ₂ As	mp-15697	6 6 3 (G)	28
AsRh ₂	mp-1102364	7 4 3 (G)	88
ZrRh	mp-2808	8 8 8 (M)	19
K ₂ S	mp-559278	5 5 4 (G)	28
Ge ₄ Rh	mp-1104286	4 4 3 (G)	135
Theoretical, metastable ($n = 60$)			
Li ₃ Ag	mp-976408	7 7 7 (G)	24
Y ₃ Co	mp-1189329	3 4 4 (G)	156
TiNi	mp-1067248	10 7 6 (G)	36
TaP	mp-1187244	9 9 9 (G)	13
Sr ₃ As ₄	mp-678007	3 3 4 (G)	62
Ga ₂ Se ₃	mp-1224809	5 5 5 (G)	30
MnN	mp-1009130	10 10 10 (G)	13
BeCl ₂	mp-1190080	3 3 3 (G)	78
MgSn	mp-1094801	8 8 6 (M)	13
TiW	mp-1216621	10 10 6 (M)	18
SrTl ₃	mp-1206636	5 5 5 (G)	32
IrCl ₃	mp-729507	3 3 6 (G)	84
Ta ₄ C ₃	mp-1218000	6 6 6 (M)	48
Sc ₂ O ₃	mp-755313	6 5 6 (G)	64
SnBr ₂	mp-978985	4 4 6 (M)	34

Table S7 (continued)

Formula	mp-id	k-point grid	NBANDS
Sn ₇ Pd ₅	mp-1219006	5 5 3 (G)	108
Tl ₄ Sn	mp-1216560	5 5 5 (G)	45
LiTl ₃	mp-1185253	6 6 6 (M)	32
PtCl ₂	mp-1219748	5 5 4 (G)	34
HCl	mp-684609	7 7 7 (G)	5
MgSn	mp-1094669	8 8 8 (M)	13
K ₅ As ₄	mp-684799	4 4 3 (G)	41
KSn ₃	mp-1185151	4 4 5 (G)	64
NbS ₂	mp-774873	10 6 2 (M)	34
YAg ₃	mp-1187811	6 6 6 (G)	37
B ₃ Ir ₂	mp-1228647	9 5 4 (G)	60
TiMo ₂	mp-1216675	5 6 14 (G)	27
Cr ₂ C	mp-1226378	10 10 7 (G)	22
ZrO ₂	mp-775909	10 6 4 (M)	36
Zn	mp-2647117	11 11 11 (G)	9
IN ₂	mp-1097011	5 5 5 (G)	24
CoSe ₂	mp-1390196	4 4 4 (G)	68
GeS	mp-12910	7 7 5 (G)	26
ZrTi	mp-1215236	9 9 6 (G)	19
VCo ₃	mp-1095057	7 6 6 (G)	72
Tl ₃ Ga	mp-1187539	5 5 5 (G)	36
CaGe ₂	mp-643542	7 7 6 (G)	23
NbRh ₂	mp-1220414	10 6 4 (M)	54
TcW	mp-1217337	10 10 6 (M)	18
MgF ₂	mp-560236	6 6 5 (G)	24
VN	mp-1017532	11 11 5 (G)	26
Na ₂ Cl	mp-990084	6 6 6 (M)	12
ZnPb ₃	mp-1187949	5 5 5 (G)	36
Sr ₃ Sn	mp-1187136	5 5 5 (G)	24
TaRe	mp-1217894	10 10 6 (M)	18
Al ₄ Si	mp-1228120	6 6 6 (M)	20
B ₆ Pb	mp-1096825	6 6 6 (M)	33
ScTc ₃	mp-867262	7 7 7 (G)	37
CrMo	mp-1226200	11 11 7 (G)	18
BPd ₅	mp-1228665	6 6 6 (M)	49
Zr ₃ Be	mp-1188016	6 6 6 (M)	35
Na ₃ Cl	mp-1064484	8 8 3 (G)	16
SiAu ₃	mp-1186998	7 7 7 (G)	31
InSe ₂	mp-1232036	7 7 3 (G)	34
TiSi ₂	mp-570991	8 8 8 (M)	17
BeAu ₃	mp-983539	7 7 7 (G)	32
Zr ₃ Ge	mp-1188039	4 4 6 (G)	78
AlSb	mp-1023918	6 6 5 (G)	16
Mg ₁₅ B	mp-1023515	4 4 3 (G)	64
HfMo	mp-1224314	10 10 6 (M)	18

E Elements and Allotropes

Every single-element structure with no matched polymorph in the dataset: the elemental references (**ELEMENT_REFERENCE**) used to decompose multi-element compounds. An experimental compound’s formula is marked with a superscript star (black if it is the stable phase, red if it is a metastable polymorph/allotrope); unmarked formulas are theoretical. Auto-generated by `report/gen_appendix_extension.py`.

Table S8: Elements and allotropes with no matched polymorph in the **extension** campaign: references used to decompose multi-element compounds. Allotropes with one or more other allotropes computed in this dataset (e.g. carbon) are grouped in Table S9 instead. Black * = stable experimental reference (thermodynamic phase); red * = metastable experimental reference; unmarked = theoretical. No Δ ICOHP column: an element is not itself a decomposition reaction.

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
Ag*	mp-124	Fm-3m	0.0021
Al*	mp-134	Fm-3m	0.0000
As*	mp-158	Cmce	0.0000
Au*	mp-81	Fm-3m	0.0000
B*	mp-160	R-3m	0.0000
Ba*	mp-122	Im-3m	0.0000
Be*	mp-87	P6 ₃ /mmc	0.0000
Br*	mp-23154	Cmce	0.0000
Ca*	mp-21	Im-3m	0.0056
Cd*	mp-94	P6 ₃ /mmc	0.0000
Cl ₂ *	mp-22848	Cmce	0.0000
Co*	mp-102	Fm-3m	0.0000
Cr*	mp-90	Im-3m	0.0000
Cs*	mp-1	Im-3m	0.0193
Cu*	mp-30	Fm-3m	0.0000
F ₂ *	mp-561203	C2/c	0.0031
Fe*	mp-13	Im-3m	0.0000
Ga*	mp-142	Cmce	0.0000
Ge*	mp-32	Fd-3m	0.0000
H ₂ *	mp-730101	P2 ₁₂ ₁₂ ₁	0.0000
Hf*	mp-103	P6 ₃ /mmc	0.0000
Hg*	mp-10861	P6/mmm	0.0030
I*	mp-23153	Cmce	0.0000
In*	mp-1055994	I4/mmm	0.0045
Ir*	mp-101	Fm-3m	0.0000
Li*	mp-1018134	R-3m	0.0000
Mg*	mp-153	P6 ₃ /mmc	0.0000
Mn*	mp-35	I-43m	0.0000
Mo*	mp-129	Im-3m	0.0000
N ₂	mp-1059834	Imma	1.1010
Nb*	mp-75	Im-3m	0.0000
Ni*	mp-23	Fm-3m	0.0000
O ₂ *	mp-1524462	C2/m	0.0000
Os*	mp-49	P6 ₃ /mmc	0.0000
P*	mp-568348	P2/c	0.0000
Pb*	mp-20483	Fm-3m	0.0000
Pd*	mp-2	Fm-3m	0.0000
Pt*	mp-126	Fm-3m	0.0000
Rb*	mp-70	Im-3m	0.0092
Re*	mp-8	P6 ₃ /mmc	0.0036
Rh*	mp-74	Fm-3m	0.0000
Ru*	mp-33	P6 ₃ /mmc	0.0000
S*	mp-77	Fddd	0.0000
Sb*	mp-104	R-3m	0.0000
Sc*	mp-67	P6 ₃ /mmc	0.0000
Se*	mp-14	P3 ₁₂₁	0.0011
Sn*	mp-623511	I4/mmm	0.0614

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
Sr*	mp-139	P6 ₃ /mmc	0.0000
Ta*	mp-50	Im-3m	0.0091
Tc*	mp-113	P6 ₃ /mmc	0.0000
Te*	mp-19	P3 ₁ 21	0.0000
Ti*	mp-46	P6 ₃ /mmc	0.0152
Tl*	mp-82	P6 ₃ /mmc	0.0000
V*	mp-146	Im-3m	0.0000
W*	mp-91	Im-3m	0.0000
Y*	mp-112	P6 ₃ /mmc	0.0000
Zn*	mp-79	P6 ₃ /mmc	0.0000
Zr*	mp-131	P6 ₃ /mmc	0.0000

F Polymorphs and Allotropes

Elements and compounds with several entries sharing a formula in the dataset, grouped here rather than scattered across the elements/compounds tables (see the table caption for detail).

Table S9: Elements (allotropes) and compounds (polymorphs) with **several entries sharing a formula** in the **extension** campaign (54 groups, 130 entries) — grouped here rather than scattered across the elements/compounds tables, since the natural comparison for a polymorph group is polymorph-to-polymorph, not the decomposition-to-elements reaction (hence no Δ ICOHP column). A rule separates each group; within a group, sorted by increasing E_{hull} . Black * = stable experimental; red * = metastable experimental; unmarked = theoretical.

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
AgN*	mp-1091417	Cmce	1.5201
AgN	mp-1009766	F-43m	1.5495
AgN	mp-1428312	Pm-3m	1.6519
AgN	mp-1424734	Ama2	1.9196
BaF ₃ *	mp-1080821	P2 ₁ /m	0.0000
BaF ₃	mp-1183303	I4/mmm	0.0612
BaO*	mp-1008500	P6 ₃ /mmc	0.0189
BaO	mp-1950989	P6 ₃ /mmc	0.0434
Be ₃ N ₂ *	mp-18337	Ia-3	0.0000
Be ₃ N ₂	mp-1017550	P-3m1	0.2081
Be ₃ P ₂ *	mp-567841	Ia-3	0.0000
Be ₃ P ₂	mp-1084771	Pn-3m	0.1567
BeCl ₂ *	mp-23267	Ibam	0.0087
BeCl ₂	mp-1172909	I-43m	3.1688
BeF ₂ *	mp-15951	P3 ₁ 21	0.0004
BeF ₂	mp-975644	P6 ₃ /mmm	2.5181
BeO	mp-1778	F-43m	0.0070
BeO*	mp-7599	P4 ₂ /mnm	0.0153
Bi ₂ O ₃ *	mp-558953	P2 ₁ /c	0.6000
Bi ₂ O ₃	mp-674644	P1	0.6199
Bi ₂ O ₃	mp-1383505	P1	0.9068
Bi ₂ O ₃	mp-714859	P1	2.4029

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
C*	mp-48	P6 ₃ /mmc	0.0031
C*	mp-66	Fd-3m	0.1123
C*	mp-47	P6 ₃ /mmc	0.1395
C	mp-1190171	Pnma	0.2927
C	mp-1080826	C2/m	0.3009
C ₃ N ₄ *	mp-571653	P-43m	0.4981
C ₃ N ₄	mp-2852	I-43d	0.5033
C ₃ N ₄	mp-1182484	P6 ₃ /m	0.5848
C ₃ N ₄	mp-563	Fd-3m	1.5119
Ca ₃ N ₂ *	mp-844	Ia-3	0.0000
Ca ₃ N ₂ *	mp-1047	R-3c	0.0167
Ca ₃ N ₂	mp-1013524	Pm-3m	0.5112
CaCl ₂ *	mp-23214	Pnnm	0.0011
CaCl ₂	mp-1120739	Fm-3m	0.0422
CaF ₂ *	mp-10464	Pnma	0.0346
CaF ₂	mp-554355	P4/mmm	0.2403
CaO*	mp-2605	Fm-3m	0.0000
CaO*	mp-1079707	Cmce	0.2098
CaO	mp-1181903	Fd-3m	2.7281
CsN ₃ *	mp-510557	I4/mcm	0.0000
CsN ₃	mp-1225892	P4/mmm	0.0113
CsO ₂ *	mp-1441	I4/mmm	0.0349
CsO ₂	mp-1096936	Cmmm	0.0940
CsP ₇ *	mp-1201790	Pbcm	0.0000
CsP ₇	mp-1213206	Pca2 ₁	0.0001
K ₂ O*	mp-971	Fm-3m	0.0000
K ₂ O	mp-1223784	P4/mmm	0.2000
KF ₃	mp-2749823	P2 ₁	0.0119
KF ₃ *	mp-1544557	P2 ₁ 2 ₁ 2 ₁	0.0125
KN ₂ *	mp-1071868	Cmc2 ₁	1.1630
KN ₂	mp-1180757	Cmcm	1.2293
KP*	mp-1058315	R-3m	1.0080
KP	mp-1057787	Immm	1.0092
Li ₂ O*	mp-1960	Fm-3m	0.0000
Li ₂ O	mp-755894	Pnma	0.0844
Li ₂ S*	mp-1125	Pnma	0.0615
Li ₂ S	mp-557142	P6 ₃ /mmc	0.1750
LiCl	mp-1185319	P6 ₃ mc	0.0039
LiCl*	mp-22905	Fm-3m	0.0243
LiF*	mp-1138	Fm-3m	0.0000
LiF	mp-1009009	Pm-3m	0.2875
LiP ₅ *	mp-2412	Pna2 ₁	0.0077
LiP ₅	mp-32760	Pna2 ₁	0.0963
MgCl ₂ *	mp-23210	R-3m	0.0000
MgCl ₂	mp-569626	Ama2	0.1991
MgF ₂ *	mp-1072956	Pnnm	0.0025
MgF ₂	mp-1062842	Amm2	0.2636

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
MgO*	mp-549706	P6 ₃ mc	0.1180
MgO	mp-1009127	Pm-3m	0.7484
MgS*	mp-1315	Fm-3m	0.0000
MgS	mp-13032	F-43m	0.0343
Na ₂ O*	mp-2352	Fm-3m	0.0000
Na ₂ O	mp-1975297	C2/m	0.0981
NaN ₃ *	mp-22003	R-3m	0.0017
NaN ₃	mp-1065265	C2/m	1.0162
NaS*	mp-409	P-62m	0.0023
NaS	mp-1180106	C2/m	1.0061
NbO*	mp-2692	Fm-3m	0.7888
NbO	mp-634965	P4/mmm	0.8240
NbP*	mp-9830	I4 ₁ /amd	0.6021
NbP	mp-1220317	Cmmm	0.6209
OsC*	mp-7142	P-6m2	0.9161
OsC	mp-999308	P6 ₃ /mmc	0.9278
OsC	mp-1009539	Fm-3m	1.3918
OsC	mp-1009537	Pm-3m	1.4795
Rb ₂ O*	mp-1394	Fm-3m	0.0032
Rb ₂ O	mp-2023482	P-3m1	0.0479
RbF	mp-11718	Fm-3m	0.0000
RbF*	mp-2064	Pm-3m	0.0516
RbN ₃ *	mp-581833	P4/mmm	0.0193
RbN ₃	mp-1219567	Amm2	0.7966
RbP*	mp-1179688	C2/m	0.9879
RbP	mp-1058499	Immm	0.9882
RbS*	mp-558071	Immm	0.0019
RbS	mp-1179684	C2/m	0.9975
ReC*	mp-1079494	P6 ₃ /mmc	0.6362
ReC	mp-1009731	Pm-3m	1.1150
RuC*	mp-10030	P-6m2	0.6494
RuC	mp-1009787	Fm-3m	0.9097
SN*	COD:7017102	P2 ₁ /c	—
SN*	mp-236	P2 ₁ /c	0.5087
SN	mp-684642	P2 ₁	0.5498
SN	mp-1173453	P2 ₁	0.7225
SN	mp-1078816	P2 ₁ /c	0.8090
SiO ₂ *	mp-9258	Pa-3	0.5770
SiO ₂	mp-10064	Fm-3m	1.2779
SiO ₂	mp-556044	P-31c	1.4574
SiO ₂	mp-638049	F4 ₁₃₂	1.7448
Sr ₃ P ₄ *	mp-14288	Fdd2	0.0000
Sr ₃ P ₄	mp-682953	P1	0.0081
SrF ₂	mp-1102911	Pnma	0.0267
SrF ₂ *	mp-1102240	Pnma	0.0646
SrO ₂ *	mp-726705	Pnma	0.0096
SrO ₂	mp-2763179	Pnma	0.0110

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
TaN*	mp-1497	P6/mmm	0.4516
TaN	mp-1009833	F-43m	0.4917
TaN	mp-1009831	Pm-3m	0.8163
TiO ₂ *	mp-1439	Pbcn	0.0045
TiO ₂ *	mp-430	P2 ₁ /c	0.0395
TiO ₂ *	mp-9173	Pnma	0.0575
TiC*	mp-567465	Fm-3m	2.1080
TiC	mp-1058887	P4/mmm	2.3757
TiN*	mp-1017982	P6 ₃ mc	0.7334
TiN	mp-1007778	F-43m	0.7337
TiN	mp-1002222	Fm-3m	0.9712
WC*	mp-13136	Fm-3m	0.4477
WC	mp-1008635	F-43m	0.6729
WC	mp-1008630	Pm-3m	0.8990

G Target Compounds

Every multi-element target compound with no matched polymorph in the dataset, with Δ ICOHP per atom and its endobondic/exobondic label where a case-1 decomposition reaction could be computed. Auto-generated from `analysis/reaction_icohp_case1.csv` and `analysis/reaction_analysis_case1_full.csv` by `report/gen_appendix_extension.py`.

Table S10: Target compounds with no matched polymorph in the **extension** campaign. Formulas with several entries in this dataset are grouped in Table S9 instead. Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. Δ ICOHP per atom, products – reactants convention (`reaction_analysis`); “–” where no case-1 reaction could be computed (missing elemental reference).

Formula	mp-id / COD	Space group	E_{hull} (eV/at)	Δ ICOHP/at. (eV)	Label
Ba ₅ P ₄ *	mp-17566	Pnma	0.0000	-0.9590	exobondic
BaCl ₂ *	mp-23199	Pnma	0.0068	-0.2244	exobondic
BaN ₂ *	mp-1102371	Cc	2.0568	-8.6248	exobondic
BaS ₃ *	mp-556296	P2 ₁ 2 ₁ 2	0.0436	-0.6233	exobondic
CaN*	mp-1058549	Fm-3m	0.3982	-4.9183	exobondic
Cs ₂ S*	mp-1077814	P6 ₃ /mmc	0.0667	0.9554	endobondic
CsCl*	mp-573697	Fm-3m	0.0071	0.8830	endobondic
CsF*	mp-8455	Pm-3m	0.1088	3.8464	endobondic
K ₂ S*	mp-1022	Fm-3m	0.0000	-0.3359	exobondic
KCl*	mp-23289	Pm-3m	0.0776	0.7616	endobondic
KN ₃	mp-636056	I4/mcm	1.0370	-5.9991	exobondic
Li ₃ N*	mp-2341	P6 ₃ /mmc	0.0039	0.6165	endobondic
Mn ₂ O ₇ *	mp-28338	P2 ₁ /c	0.3346	-1.4687	exobondic
MnO ₂ *	mp-510408	P4 ₂ /nmn	0.0464	-0.5274	exobondic
Na ₂ Cl	mp-1077015	Cmmm	0.2391	-0.3965	exobondic
PbN ₆ *	mp-667338	Pnma	0.1093	-2.5257	exobondic
RbCl*	mp-23299	Pm-3m	0.0298	1.7924	endobondic
S ₂ N*	COD:4031496	P4 ₂ nm	–	-0.9767	exobondic
SrN*	mp-1058586	Fm-3m	0.4343	-4.5395	exobondic
SrS*	mp-10627	Pm-3m	0.2996	-0.1088	exobondic

H Endobondic Reactions

The subset of case-1 decomposition reactions ($A_a B_b \rightarrow a A + b B$, standard-state elemental references, balanced by `reaction_icohp.py`) labeled **endobondic** ($\Delta\text{ICOHP} \geq 0$, products – reactants convention — note this is the sign *opposite* `reaction_icohp.py`’s own `delta_icohp_per_atom` column reported elsewhere in this project; see `delta.py`’s docstring).

Table S11: Decomposition-into-elements (case 1) reactions labeled **endobondic** ($\Delta\text{ICOHP} \geq 0$, products – reactants) among the 208 compounds of the **extension** campaign. * = experimental starting compound (unmarked = theoretical).

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	$\Delta\text{ICOHP/at.}$ (eV)
0.3333 Be ₃ N ₂ $\rightarrow$ Be + 0.3333 N ₂	mp-1017550	0.2081	0.3960
BeCl ₂ * $\rightarrow$ Be + Cl ₂	mp-23267	0.0087	1.8204
BeF ₂ * $\rightarrow$ Be + F ₂	mp-15951	0.0004	3.1514
BeO* $\rightarrow$ Be + 0.5 O ₂	mp-7599	0.0153	2.7864
BeO $\rightarrow$ Be + 0.5 O ₂	mp-1778	0.0070	4.4391
0.3333 C ₃ N ₄ * $\rightarrow$ C + 0.6667 N ₂	mp-571653	0.4981	1.1208
0.3333 C ₃ N ₄ $\rightarrow$ C + 0.6667 N ₂	mp-2852	0.5033	1.8456
0.5 Cs ₂ S* $\rightarrow$ Cs + 0.5 S	mp-1077814	0.0667	0.9554
CsCl* $\rightarrow$ Cs + 0.5 Cl ₂	mp-573697	0.0071	0.8830
CsF* $\rightarrow$ Cs + 0.5 F ₂	mp-8455	0.1088	3.8464
CsO ₂ * $\rightarrow$ Cs + O ₂	mp-1441	0.0349	0.0666
CsO ₂ $\rightarrow$ Cs + O ₂	mp-1096936	0.0940	0.7743
KCl* $\rightarrow$ K + 0.5 Cl ₂	mp-23289	0.0776	0.7616
0.5 Li ₂ O* $\rightarrow$ Li + 0.25 O ₂	mp-1960	0.0000	2.3366
0.5 Li ₂ O $\rightarrow$ Li + 0.25 O ₂	mp-755894	0.0844	1.9481
0.5 Li ₂ S* $\rightarrow$ Li + 0.5 S	mp-1125	0.0615	1.8505
0.5 Li ₂ S $\rightarrow$ Li + 0.5 S	mp-557142	0.1750	1.5052
0.3333 Li ₃ N* $\rightarrow$ Li + 0.1667 N ₂	mp-2341	0.0039	0.6165
LiCl* $\rightarrow$ Li + 0.5 Cl ₂	mp-22905	0.0243	2.8030
LiCl $\rightarrow$ Li + 0.5 Cl ₂	mp-1185319	0.0039	2.5005
LiF* $\rightarrow$ Li + 0.5 F ₂	mp-1138	0.0000	4.4673
LiF $\rightarrow$ Li + 0.5 F ₂	mp-1009009	0.2875	4.5131
LiP ₅ * $\rightarrow$ Li + 5 P	mp-2412	0.0077	0.9043
LiP ₅ $\rightarrow$ Li + 5 P	mp-32760	0.0963	0.8692
MgCl ₂ * $\rightarrow$ Mg + Cl ₂	mp-23210	0.0000	0.1568
MgF ₂ * $\rightarrow$ Mg + F ₂	mp-1072956	0.0025	0.0015
MgF ₂ $\rightarrow$ Mg + F ₂	mp-1062842	0.2636	0.0229
0.5 Rb ₂ O* $\rightarrow$ Rb + 0.25 O ₂	mp-1394	0.0032	0.7235
RbCl* $\rightarrow$ Rb + 0.5 Cl ₂	mp-23299	0.0298	1.7924
RbF* $\rightarrow$ Rb + 0.5 F ₂	mp-2064	0.0516	3.9244
RbF $\rightarrow$ Rb + 0.5 F ₂	mp-11718	0.0000	1.9497
SiO ₂ * $\rightarrow$ Si + O ₂	mp-9258	0.5770	3.3332
SiO ₂ $\rightarrow$ Si + O ₂	mp-10064	1.2779	2.1735
SiO ₂ $\rightarrow$ Si + O ₂	mp-556044	1.4574	0.0958
SiO ₂ $\rightarrow$ Si + O ₂	mp-638049	1.7448	0.3849

I Exobondic Reactions

The complementary **exobondic** subset ($\Delta\text{ICOHP} < 0$), same convention and provenance as Appendix H.

Table S12: Decomposition-into-elements (case 1) reactions labeled **exobondic** ($\Delta\text{ICOHP} < 0$, products – reactants) among the 208 compounds of the **extension** campaign. * = experimental starting compound (unmarked = theoretical).

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	$\Delta\text{ICOHP/at.}$ (eV)
$\text{AgN}^* \rightarrow \text{Ag} + 0.5 \text{ N}_2$	mp-1091417	1.5201	-7.5875
$\text{AgN} \rightarrow \text{Ag} + 0.5 \text{ N}_2$	mp-1009766	1.5495	-6.4957
$\text{AgN} \rightarrow \text{Ag} + 0.5 \text{ N}_2$	mp-1424734	1.9196	-8.0907
$\text{AgN} \rightarrow \text{Ag} + 0.5 \text{ N}_2$	mp-1428312	1.6519	-5.3559
$0.2 \text{ Ba}_5\text{P}_4^* \rightarrow \text{Ba} + 0.8 \text{ P}$	mp-17566	0.0000	-0.9590
$\text{BaCl}_2^* \rightarrow \text{Ba} + \text{Cl}_2$	mp-23199	0.0068	-0.2244
$\text{BaF}_3^* \rightarrow \text{Ba} + 1.5 \text{ F}_2$	mp-1080821	0.0000	-0.8680
$\text{BaF}_3 \rightarrow \text{Ba} + 1.5 \text{ F}_2$	mp-1183303	0.0612	-0.2493
$\text{BaN}_2^* \rightarrow \text{Ba} + \text{N}_2$	mp-1102371	2.0568	-8.6248
$\text{BaO}^* \rightarrow \text{Ba} + 0.5 \text{ O}_2$	mp-1008500	0.0189	-0.3221
$\text{BaO} \rightarrow \text{Ba} + 0.5 \text{ O}_2$	mp-1950989	0.0434	-1.6129
$\text{BaS}_3^* \rightarrow \text{Ba} + 3 \text{ S}$	mp-556296	0.0436	-0.6233
$0.3333 \text{ Be}_3\text{N}_2^* \rightarrow \text{Be} + 0.3333 \text{ N}_2$	mp-18337	0.0000	-0.0746
$0.3333 \text{ Be}_3\text{P}_2^* \rightarrow \text{Be} + 0.6667 \text{ P}$	mp-567841	0.0000	-0.2111
$0.3333 \text{ Be}_3\text{P}_2 \rightarrow \text{Be} + 0.6667 \text{ P}$	mp-1084771	0.1567	-0.1795
$\text{BeCl}_2 \rightarrow \text{Be} + \text{Cl}_2$	mp-1172909	3.1688	-5.7807
$\text{BeF}_2 \rightarrow \text{Be} + \text{F}_2$	mp-975644	2.5181	-0.6816
$0.3333 \text{ C}_3\text{N}_4 \rightarrow \text{C} + 0.6667 \text{ N}_2$	mp-1182484	0.5848	-3.6399
$0.3333 \text{ C}_3\text{N}_4 \rightarrow \text{C} + 0.6667 \text{ N}_2$	mp-563	1.5119	-0.4498
$0.3333 \text{ Ca}_3\text{N}_2^* \rightarrow \text{Ca} + 0.3333 \text{ N}_2$	mp-1047	0.0167	-4.4881
$0.3333 \text{ Ca}_3\text{N}_2^* \rightarrow \text{Ca} + 0.3333 \text{ N}_2$	mp-844	0.0000	-4.4374
$0.3333 \text{ Ca}_3\text{N}_2 \rightarrow \text{Ca} + 0.3333 \text{ N}_2$	mp-1013524	0.5112	-4.4898
$\text{CaCl}_2^* \rightarrow \text{Ca} + \text{Cl}_2$	mp-23214	0.0011	-0.3199
$\text{CaCl}_2 \rightarrow \text{Ca} + \text{Cl}_2$	mp-1120739	0.0422	-0.2208
$\text{CaF}_2^* \rightarrow \text{Ca} + \text{F}_2$	mp-10464	0.0346	-0.2227
$\text{CaF}_2 \rightarrow \text{Ca} + \text{F}_2$	mp-554355	0.2403	-0.1814
$\text{CaN}^* \rightarrow \text{Ca} + 0.5 \text{ N}_2$	mp-1058549	0.3982	-4.9183
$\text{CaO}^* \rightarrow \text{Ca} + 0.5 \text{ O}_2$	mp-1079707	0.2098	-2.2127
$\text{CaO}^* \rightarrow \text{Ca} + 0.5 \text{ O}_2$	mp-2605	0.0000	-0.9525
$\text{CaO} \rightarrow \text{Ca} + 0.5 \text{ O}_2$	mp-1181903	2.7281	-2.9971
$\text{CsN}_3^* \rightarrow \text{Cs} + 1.5 \text{ N}_2$	mp-510557	0.0000	-2.2903
$\text{CsN}_3 \rightarrow \text{Cs} + 1.5 \text{ N}_2$	mp-1225892	0.0113	-1.6664
$\text{CsP}_7^* \rightarrow \text{Cs} + 7 \text{ P}$	mp-1201790	0.0000	-0.2978
$\text{CsP}_7 \rightarrow \text{Cs} + 7 \text{ P}$	mp-1213206	0.0001	-0.2667
$0.5 \text{ K}_2\text{O}^* \rightarrow \text{K} + 0.25 \text{ O}_2$	mp-971	0.0000	-0.4357
$0.5 \text{ K}_2\text{O} \rightarrow \text{K} + 0.25 \text{ O}_2$	mp-1223784	0.2000	-1.5638
$0.5 \text{ K}_2\text{S}^* \rightarrow \text{K} + 0.5 \text{ S}$	mp-1022	0.0000	-0.3359
$\text{KF}_3^* \rightarrow \text{K} + 1.5 \text{ F}_2$	mp-1544557	0.0125	-1.1511
$\text{KF}_3 \rightarrow \text{K} + 1.5 \text{ F}_2$	mp-2749823	0.0119	-1.1505
$\text{KN}_2^* \rightarrow \text{K} + \text{N}_2$	mp-1071868	1.1630	-3.4244
$\text{KN}_2 \rightarrow \text{K} + \text{N}_2$	mp-1180757	1.2293	-4.2865
$\text{KN}_3 \rightarrow \text{K} + 1.5 \text{ N}_2$	mp-636056	1.0370	-5.9991
$\text{KP}^* \rightarrow \text{K} + \text{P}$	mp-1058315	1.0080	-2.1600
$\text{KP} \rightarrow \text{K} + \text{P}$	mp-1057787	1.0092	-2.1366
$\text{MgCl}_2 \rightarrow \text{Mg} + \text{Cl}_2$	mp-569626	0.1991	-0.0862
$\text{MgO}^* \rightarrow \text{Mg} + 0.5 \text{ O}_2$	mp-549706	0.1180	-1.4179
$\text{MgO} \rightarrow \text{Mg} + 0.5 \text{ O}_2$	mp-1009127	0.7484	-1.0763
$\text{MgS}^* \rightarrow \text{Mg} + \text{S}$	mp-1315	0.0000	-0.3401
$\text{MgS} \rightarrow \text{Mg} + \text{S}$	mp-13032	0.0343	-0.4163
$0.5 \text{ Mn}_2\text{O}_7^* \rightarrow \text{Mn} + 1.75 \text{ O}_2$	mp-28338	0.3346	-1.4687
$\text{MnO}_2^* \rightarrow \text{Mn} + \text{O}_2$	mp-510408	0.0464	-0.5274

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	Δ ICOHP/at. (eV)
$0.5 \text{ Na}_2\text{Cl} \rightarrow \text{Na} + 0.25 \text{ Cl}_2$	mp-1077015	0.2391	-0.3965
$0.5 \text{ Na}_2\text{O}^* \rightarrow \text{Na} + 0.25 \text{ O}_2$	mp-2352	0.0000	-1.3057
$0.5 \text{ Na}_2\text{O} \rightarrow \text{Na} + 0.25 \text{ O}_2$	mp-1975297	0.0981	-1.5619
$\text{NaN}_3^* \rightarrow \text{Na} + 1.5 \text{ N}_2$	mp-22003	0.0017	-1.5201
$\text{NaN}_3 \rightarrow \text{Na} + 1.5 \text{ N}_2$	mp-1065265	1.0162	-5.4248
$\text{NaS}^* \rightarrow \text{Na} + \text{S}$	mp-409	0.0023	-0.4911
$\text{NaS} \rightarrow \text{Na} + \text{S}$	mp-1180106	1.0061	-2.3999
$\text{NbO}^* \rightarrow \text{Nb} + 0.5 \text{ O}_2$	mp-2692	0.7888	-2.3179
$\text{NbO} \rightarrow \text{Nb} + 0.5 \text{ O}_2$	mp-634965	0.8240	-3.1495
$\text{NbP}^* \rightarrow \text{Nb} + \text{P}$	mp-9830	0.6021	-2.4724
$\text{NbP} \rightarrow \text{Nb} + \text{P}$	mp-1220317	0.6209	-1.1731
$\text{OsC}^* \rightarrow \text{Os} + \text{C}$	mp-7142	0.9161	-2.0168
$\text{OsC} \rightarrow \text{Os} + \text{C}$	mp-1009537	1.4795	-2.3184
$\text{OsC} \rightarrow \text{Os} + \text{C}$	mp-1009539	1.3918	-3.1141
$\text{OsC} \rightarrow \text{Os} + \text{C}$	mp-999308	0.9278	-2.9572
$\text{PbN}_6^* \rightarrow \text{Pb} + 3 \text{ N}_2$	mp-667338	0.1093	-2.5257
$0.5 \text{ Rb}_2\text{O} \rightarrow \text{Rb} + 0.25 \text{ O}_2$	mp-2023482	0.0479	-0.7592
$\text{RbN}_3^* \rightarrow \text{Rb} + 1.5 \text{ N}_2$	mp-581833	0.0193	-1.3478
$\text{RbN}_3 \rightarrow \text{Rb} + 1.5 \text{ N}_2$	mp-1219567	0.7966	-3.5636
$\text{RbP}^* \rightarrow \text{Rb} + \text{P}$	mp-1179688	0.9879	-1.9315
$\text{RbP} \rightarrow \text{Rb} + \text{P}$	mp-1058499	0.9882	-1.9104
$\text{RbS}^* \rightarrow \text{Rb} + \text{S}$	mp-558071	0.0019	-0.3217
$\text{RbS} \rightarrow \text{Rb} + \text{S}$	mp-1179684	0.9975	-2.7347
$\text{ReC}^* \rightarrow \text{Re} + \text{C}$	mp-1079494	0.6362	-2.6625
$\text{ReC} \rightarrow \text{Re} + \text{C}$	mp-1009731	1.1150	-1.9955
$\text{RuC}^* \rightarrow \text{Ru} + \text{C}$	mp-10030	0.6494	-2.1301
$\text{RuC} \rightarrow \text{Ru} + \text{C}$	mp-1009787	0.9097	-2.8605
$0.5 \text{ S}_2\text{N}^* \rightarrow \text{S} + 0.25 \text{ N}_2$	COD:4031496	—	-0.9767
$\text{SN}^* \rightarrow \text{S} + 0.5 \text{ N}_2$	COD:7017102	—	-1.3963
$\text{SN}^* \rightarrow \text{S} + 0.5 \text{ N}_2$	mp-236	0.5087	-2.3179
$\text{SN} \rightarrow \text{S} + 0.5 \text{ N}_2$	mp-1078816	0.8090	-3.7082
$\text{SN} \rightarrow \text{S} + 0.5 \text{ N}_2$	mp-1173453	0.7225	-3.1673
$\text{SN} \rightarrow \text{S} + 0.5 \text{ N}_2$	mp-684642	0.5498	-1.7362
$0.3333 \text{ Sr}_3\text{P}_4^* \rightarrow \text{Sr} + 1.333 \text{ P}$	mp-14288	0.0000	-1.0472
$0.3333 \text{ Sr}_3\text{P}_4 \rightarrow \text{Sr} + 1.333 \text{ P}$	mp-682953	0.0081	-1.0793
$\text{SrF}_2^* \rightarrow \text{Sr} + \text{F}_2$	mp-1102240	0.0646	-0.5421
$\text{SrF}_2 \rightarrow \text{Sr} + \text{F}_2$	mp-1102911	0.0267	-0.1481
$\text{SrN}^* \rightarrow \text{Sr} + 0.5 \text{ N}_2$	mp-1058586	0.4343	-4.5395
$\text{SrO}_2^* \rightarrow \text{Sr} + \text{O}_2$	mp-726705	0.0096	-1.9205
$\text{SrO}_2 \rightarrow \text{Sr} + \text{O}_2$	mp-2763179	0.0110	-1.7522
$\text{SrS}^* \rightarrow \text{Sr} + \text{S}$	mp-10627	0.2996	-0.1088
$\text{TaN}^* \rightarrow \text{Ta} + 0.5 \text{ N}_2$	mp-1497	0.4516	-5.9309
$\text{TaN} \rightarrow \text{Ta} + 0.5 \text{ N}_2$	mp-1009831	0.8163	-4.0540
$\text{TaN} \rightarrow \text{Ta} + 0.5 \text{ N}_2$	mp-1009833	0.4917	-5.3385
$\text{TiO}_2^* \rightarrow \text{Ti} + \text{O}_2$	mp-1439	0.0045	-0.6461
$\text{TiO}_2^* \rightarrow \text{Ti} + \text{O}_2$	mp-9173	0.0575	-0.6509
$\text{TiO}_2^* \rightarrow \text{Ti} + \text{O}_2$	mp-430	0.0395	-0.6267
$\text{TlC}^* \rightarrow \text{Tl} + \text{C}$	mp-567465	2.1080	-3.1404
$\text{TlC} \rightarrow \text{Tl} + \text{C}$	mp-1058887	2.3757	-3.3638
$\text{TlN}^* \rightarrow \text{Tl} + 0.5 \text{ N}_2$	mp-1017982	0.7334	-3.7498
$\text{TlN} \rightarrow \text{Tl} + 0.5 \text{ N}_2$	mp-1002222	0.9712	-2.3228
$\text{TlN} \rightarrow \text{Tl} + 0.5 \text{ N}_2$	mp-1007778	0.7337	-3.1948
$\text{WC}^* \rightarrow \text{W} + \text{C}$	mp-13136	0.4477	-6.2684
$\text{WC} \rightarrow \text{W} + \text{C}$	mp-1008630	0.8990	-5.9400
$\text{WC} \rightarrow \text{W} + \text{C}$	mp-1008635	0.6729	-7.7315

J ICOHP and ICOBI Antibonding Descriptors

J.1 Implementation

The ICOHP antibonding-population-near-frontier descriptor (§6.2, `cohp_extraction.py`) is extended here to ICOBI: same one-sided window ($E_{\text{ref}} - \Delta E$, E_{ref}], same E_{ref} (Fermi level for metals, VBM for gapped compounds — logic entirely independent of which curve is integrated), read this time from `COBICAR.lobster/ICOBILIST.lobster` instead of `COHPCAR.lobster/ICOHPLIST.lobster` (`pymatgen.io.lobster.outputs.Cohpcar(are_cobis=True)`).

Critical point, empirically verified, not assumed: ICOBI does **not** share ICOHP’s sign convention. Whereas "negative ICOHP = bonding" (§2.4), ICOBI follows the opposite convention, "positive ICOBI = bonding" (like COOP, not like COHP). Verified two ways on real data: (1) exact cross-check against `ICOBILIST.lobster` (max deviation 0.0 across 66 bond labels, `tests/test_cohp_extraction.py`) — `pymatgen` does not silently renormalize the sign when switching from COHP to COBI parsing; (2) on `extension_SiO2_anchor`’s strongest Si–O bond (ICOHP = -5.599 , ICOBI = $+0.513$), the deep bonding region (-10 to -3 eV, far from any frontier effects) gives a mean COHP(E) of -0.338 but a mean COBI(E) of $+0.041$ — opposite signs in the same physically bonding-dominated region. ICOBI’s antibonding part is therefore its *negative* part, not its positive part – the clip direction (`integrate_icobi_antibonding_in_window()`) is deliberately the mirror image of the ICOHP version, not a copy.

Computed across the 394 compounds with a `COBICAR.lobster` file (`analysis/compute_icobi_antibonding_all.py`); 392 with `is_metal` available.

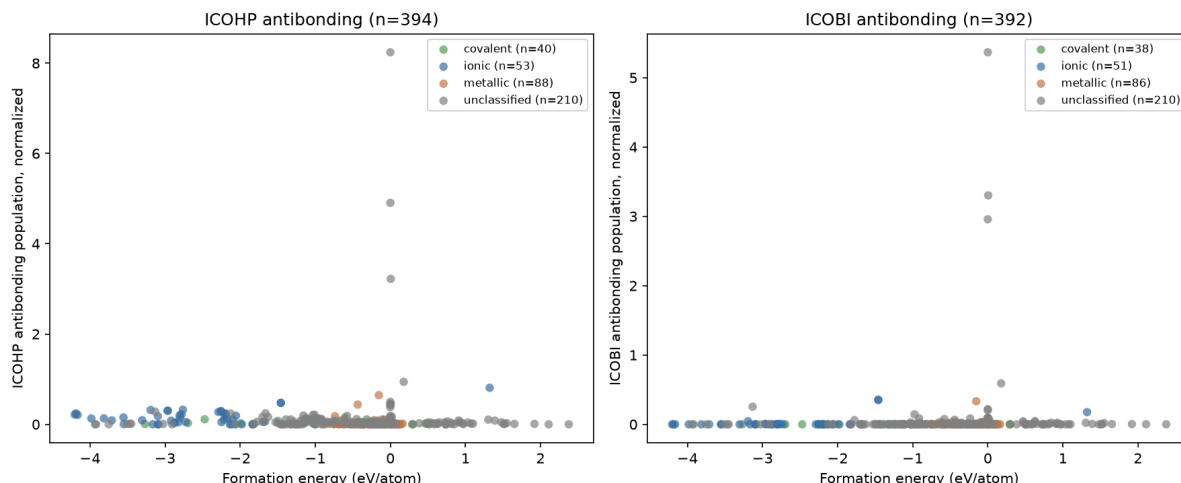
J.2 Correlation with formation energy

Table S13: Spearman correlations (normalized, $\Delta E = 1.0$ eV) against `formation_energy_per_atom` — ICOHP (n=394) and ICOBI (n=392) antibonding, same population.

Group	n (ICOHP)	ρ (ICOHP)	n (ICOBI)	ρ (ICOBI)
all	391	-0.2473^{***}	385	$+0.2518^{***}$
covalent	40	-0.4781^{**}	38	-0.1755
ionic	53	-0.1842	51	$+0.1289$
metallic	88	-0.1864	86	$+0.0011$
<code>is_metal=False</code>	166	-0.2621^{***}	162	$+0.0661$
<code>is_metal=True</code>	225	-0.0154	223	$+0.1263$

$^{**}p < 0.01$; $^{***}p < 0.001$; no star = not significant. Full detail (raw + normalized, all three ΔE):

`analysis/stats_summary_antibonding_joint.json`.



The two descriptors behave differently. ICOBI antibonding’s global sign is the *opposite* of ICOHP antibonding’s (+0.25 vs. −0.25) — for ICOHP, more occupied antibonding character near E_F is associated with *more* stable formation (already noted as counterintuitive in §6.2); for ICOBI, the relationship runs the naively expected way (more antibonding $\Rightarrow$ less stable). But **no ICOBI subgroup survives stratification** (covalent, ionic, metallic, `is_metal`: $p > 0.05$ throughout) — unlike ICOHP, whose covalent subgroup ($n = 40$, $\rho = -0.478$) remains the project’s most robust stratified result. ICOBI’s global correlation is therefore, like reaction ICOHP’s (§6.2), likely a between-bond-type clustering effect rather than a robust relationship within a homogeneous group.

J.3 $\Delta(\text{antibonding})$ of the decomposition-into-elements reaction

For a case-1 reaction (compound $\rightarrow$ elements), $\Delta(\text{antibonding})$ is **not** computed by `reaction_analysis/delta.py` (which combines *extensive* per-formula-unit ICOHP/ICOBI sums with integer coefficients): the antibonding population near the frontier is built from LOBSTER’s own “average” trace, an *intensive* quantity (averaged over all bonds, not summed) with no natural extensive decomposition. Convention chosen here, documented as an explicit methodological choice rather than a silent reuse: $\Delta = (\text{atomic-fraction-weighted average of the elements’ own antibonding descriptors}) - (\text{the compound’s own antibonding descriptor})$, still in products − reactants convention. Computed for 320 case-1 reactions (`analysis/compute_delta_antibonding_case1.py`).

Table S14: Spearman correlations, $\Delta(\text{antibonding})$ of the decomposition reaction (case 1, $n = 316\text{--}320$).

Descriptor	Target	n	ρ
$\Delta(\text{ICOHP antibonding})$	<code>formation_energy_per_atom</code>	316	−0.6439***
$\Delta(\text{ICOBI antibonding})$	<code>formation_energy_per_atom</code>	316	−0.4457***
$\Delta(\text{ICOHP antibonding})$	<code>energy_above_hull</code>	320	−0.1868***
$\Delta(\text{ICOBI antibonding})$	<code>energy_above_hull</code>	320	−0.1524**

** $p < 0.01$; *** $p < 0.001$.

$\Delta(\text{ICOHP antibonding})$ is, at this stage, the **strongest continuous correlation in the whole project** ($\rho = -0.644$, stronger than reaction ICOHP in §6.2, $\rho = 0.295$). Notable: unlike the raw values (opposite signs, above), **both Δ s run the same direction** (both negative) — the delta construction appears to align the two descriptors on a consistent reading, where their

raw values do not. The two Δ s are themselves correlated with each other ($\rho = 0.532$, $p < 0.001$, $n = 320$): related but not redundant.

J.4 Stratification by bond type

Table S15: Spearman correlations, $\Delta(\text{antibonding})$ against `formation_energy_per_atom`, by subgroup (case 1, $n = 316$).

Subgroup	n	ρ (ΔICOHP)	n	ρ (ΔICOB)
all	316	−0.6439***	316	−0.4457***
covalent	29	−0.5798***	29	−0.2044
ionic	51	−0.4205**	51	+0.0608
metallic	86	−0.4858***	86	−0.4872***
is_metal=False	145	−0.6210***	145	−0.3248***
is_metal=True	171	−0.5490***	171	−0.4653***

** $p < 0.01$; *** $p < 0.001$; no star = not significant.

$\Delta(\text{ICOHP antibonding})$ is, to date, the first descriptor in the whole project whose global correlation survives *every* stratification tested — covalent, ionic, metallic, `is_metal` on both sides, all significant at $p < 0.01$ or better. No other descriptor in this report manages that: the raw antibonding population (§6.2) keeps only its covalent subgroup, reaction ICOHP (§6.2) loses every `bond_type` stratum and survives only on `is_metal`. $\Delta(\text{ICOB antibonding})$ is more mixed: survives on metallic and both `is_metal` sides but not on covalent ($p = 0.29$) or ionic ($p = 0.67$). This remains a newly chosen construction (previous subsection) — its magnitude will need to reproduce on an independent dataset before being treated as a fully established result, but surviving full stratification makes it, as it stands, the project’s most robust result.

J.5 Reaction $\Delta(\text{ICOHP})$ restricted to a window near E_F

Extension proposed in the Conclusion (§6.2): rather than ICOHP integrated over the whole occupied range (§6.2) or only the antibonding part near E_F (previous subsections), a third construction — the *signed* ICOHP (bonding *and* antibonding, not clipped) but integrated *only* over the one-sided window $(E_F - \Delta E, E_F]$, $\Delta E = 1.0$ eV, summed over *every* bond label (extensive, like the full ICOHP sum) rather than just the "average" trace — then combined with the same reaction stoichiometry as the full $\Delta(\text{ICOHP})$ (`reaction_icohp.py`, reactant — predicted products convention). Implementation: `cohp_extraction.icohp_windowed_per_atom()` reads the difference between two points on the *cumulative* ICOHP curve LOBSTER already computes in `COHPCAR.lobster` (`ICOHP_cumulative( $E_F$ ) - ICOHP_cumulative( $E_F - \Delta E$ )`, per label), no new trapezoidal integration — validated: at very large ΔE (covering the whole calculation window), this quantity reproduces the full ICOHP sum to within $\sim 0.1\%$.

An intermediate result between the other two constructions: unlike the full reaction ICOHP (§6.2), which loses *every* `bond_type` stratum, this windowed version **survives on the ionic subgroup** ($\rho = 0.524$, $p < 0.001$) and on `is_metal` on both sides — but not on metallic, and only marginally on covalent ($p = 0.053$). Less robust than $\Delta(\text{ICOHP antibonding})$ (previous subsection, survives everywhere), more robust than the full reaction ICOHP. Cross-correlation with $\Delta(\text{ICOHP antibonding})$: $\rho = -0.599$ ($p < 0.001$, $n = 320$) — strongly *anticorrelated* despite their closely related construction (same window, same E_F), a sign that clipping to the antibonding-only part (previous subsection) captures qualitatively different information from the full signed ICOHP in the same window. Cross-correlation with the full reaction ICOHP:

Table S16: Spearman correlations, $\Delta(\text{windowed ICOHP near } E_F)$ against `formation_energy_per_atom` (case 1, $n = 316$).

Subgroup	n	ρ
all	316	+0.3712***
covalent	29	+0.3635 (p=0.053)
ionic	51	+0.5243***
metallic	86	+0.1721
is_metal=False	145	+0.4660***
is_metal=True	171	+0.1848*

* $p < 0.05$; *** $p < 0.001$; no star = not significant. Against `energy_above_hull`, no significant correlation anywhere ($n = 320$, $p > 0.07$ throughout) — the same behavior as every other ICOHP/ICOBI descriptor in this report.

$\rho = 0.231$ ($p < 0.001$) — related but clearly distinct, not simply a restriction of the same quantity.

K Reitz & Dronskowski reaction structures

The §6.2 validation (Table 3) only compares the arithmetic of the manuscript’s own published ICOHP values — independent of any local crystal structure. The table below documents, for each of the 16 species involved in these 7 reactions, the structure as described by the manuscript (where it specifies one), the Materials Project/COD entry used in this dataset, and whether an independent verification was possible. Three cases revealed a real bug or ambiguity, since fixed: N_2 (a theoretical polymeric phase substituted for an isolated gaseous dimer, §6.2), $\text{Pb}(\text{N}_3)_2$ (the wrong Pnma polymorph substituted for the Pna2_1 the manuscript cites), and ZnSn ’s competing tin phase (β -tin substituted for α -tin).

Table S17: The 16 species across Reitz & Dronskowski’s 7 reactions (ic-2026-04181q, Table 3): structure as described by the manuscript, the MP/COD entry used in this dataset, and verification status.

Species (role)	Manuscript structure	Our source	Verification
$\text{Pb}(\text{N}_3)_2$ (target)	lead azide, ICSD 28336/16887 (Glen 1963)	mp-620058, Pna2_1, $E_{\text{hull}} = 0.110$	Yes — exact polymorph identified 2026-08-17, substituted for mp-667338 (Pnma) already present in the dataset
Pb (element)	not specified (standard elemental reference)	mp-20483, Fm-3m, $E_{\text{hull}} = 0.000$, on-hull	not required — no polymorph ambiguity for fcc metallic Pb
N_2 (element, 4 reactions)	gaseous $\text{N}\equiv\text{N}$, $d \approx 1.10 \text{ \AA}$	gasref_N2_dimerbox, hand-built isolated dimer ($10\times 10\times 10 \text{ \AA}$ box), $d(\text{N-N}) = 1.113 \text{ \AA}$, ICOHP = -22.998 eV	Yes — within 0.7% of the manuscript’s value (-23.161 eV). Before 2026-08-17: mp-1059834 (theoretical, polymeric phase, $\text{N-N} = 1.296 \text{ \AA}$) — a real bug, fixed
S_4N_2 (target)	not specified beyond composition/ICOHP	COD 4031496, P4_2nm ($Z = 4$), no MP entry exists for this composition	no — COD experimental structure used as-is, no independent confirmation it is the manuscript’s exact polymorph

Species (role)	Manuscript structure	Our source	Verification
S ₈ (element)	not specified	mp-77, Fddd (α -S ₈), $E_{hull} = 0.000$, on-hull	not required — unambiguous standard reference
S ₄ N ₄ (target)	not specified	COD 7017102, P2_1/c ($Z = 4$)	no
ZnSn (target)	implicit: 6 Zn-Sn bonds + 1 Zn-Zn bond/formula unit (consistent with a NiAs prototype)	gasref_ZnSn_NiAs, hand-built, P6_3/mmc (#194), no MP/COD entry (no stable real Zn-Sn intermetallic exists)	partial — the (6+1) coordination is consistent with NiAs, but no independent crystallographic reference was found to confirm the manuscript’s exact lattice parameters
Zn (element)	not specified	mp-79, P6_3/mmc, $E_{hull} = 0.000$, on-hull	not required
Sn (element, ZnSn’s competing phase)	implies a phase distinct from the dataset’s “default” β -tin	mp-117, Fd-3m (α -tin, diamond-cubic), substituted for mp-623511 (β -tin, I4/mmm)	partial — α -tin (the low-temperature fcc form) is the structurally consistent competing phase for ZnSn; no independent confirmation the manuscript uses mp-117 specifically
CaO[sphalerite] (target)	F-43m (#216), explicitly cited	gasref_Ca0_sphalerite, hand-built (no MP entry for this metastable CaO polymorph), F-43m confirmed by symmetry	Yes (space group)
CaO[rocksalt] (product)	Fm-3m (#225), explicitly cited	mp-2605, Fm-3m, $E_{hull} = 0.000$, on-hull	Yes
CaN (target)	not specified	mp-1058549, Fm-3m (rocksalt-type), $E_{hull} = 0.398$	no
Ca ₃ N ₂ (product)	not specified	mp-844, Ia-3 (anti-bixbyite), $E_{hull} = 0.000$, on-hull	no — the only on-hull entry, the most defensible default choice
Mn ₂ O ₇ (target)	not specified	mp-28338, P2_1/c, $E_{hull} = 0.335$	no
MnO ₂ (product)	not specified	mp-510408, P4_2/mnm (rutile/pyrolusite)	no
O ₂ (element)	gaseous O ₂ , triplet ground state	mp-1524462, C2/m, $E_{hull} = 0.000$, on-hull, dedicated spin-polarized treatment	structural — isolated dimer $d(\text{O-O}) = 1.222 \text{ \AA}$ vs. $1.208 \text{ \AA}$ literature (within 1%)

N₂ reproduces the manuscript’s value tightly (0.7%). ZnSn does not: the two elemental references (Zn, Sn) each diverge by $\sim 37\text{--}38\%$, while the ZnSn compound itself is much closer (6.6%) — because the reaction’s ΔICOHP is the small difference of large numbers, this compounds into a $\Delta\text{ICOHP} \sim 2.9\times$ larger in magnitude than the manuscript’s, even though the sign and qualitative conclusion (bonding-favorable decomposition) match. This is consistent with a bulk intermetallic’s ICOHP sum being far more sensitive to basis/pseudopotential/cutoff choices than a simple gas dimer, not with an implementation bug (§6.2’s 7-case validation, which uses the manuscript’s own numbers throughout, reproduces those

Table S18: This project’s own DFT+LOBSTER ICOHP values against the manuscript’s transcribed values, for the two species where a local calculation was run (N₂: `gasref_N2_dimerbox`; ZnSn: `gasref_ZnSn_NiAs` decomposing against the dataset’s default β -Sn reference, `extension_Sn_mp-623511`) — unlike Table 3, these are independent VASP+LOBSTER runs, not a reconstruction of the manuscript’s own numbers.

Species / reaction	Manuscript (eV)	This project’s DFT (eV)	Difference
N ₂ (element)	−23.161	−22.998	0.7%
ZnSn (reactant)	−16.869	−17.991	6.6%
Zn (product)	−12.686	−17.402	37.2%
Sn (product, β)	−7.672	−10.617	38.4%
Δ ICOHP, ZnSn→Zn+Sn	−3.489 eV (−337 kJ/mol)	−10.027 eV (−968 kJ/mol)	2.9×

numbers essentially exactly).

Reading key. “Verification: Yes” means the structure was confirmed to match the manuscript’s explicit description (a cited space group, or a recomputed bond distance compared directly). “Partial” means the geometry/chemistry is consistent with what the manuscript describes but no independent crystallographic citation allowed a strict confirmation. “No” means only the composition was verified: the manuscript does not specify a space group for that species, so the most reasonable MP/COD entry was used with no guarantee it is *the* exact structure the authors used. This does not invalidate the §6.2 validation (which only compares the arithmetic reconstruction of the manuscript’s own published ICOHP values, independent of any local structure) but it does bound the scope of any future comparison between our own DFT+LOBSTER calculations and the manuscript’s numbers for these species.